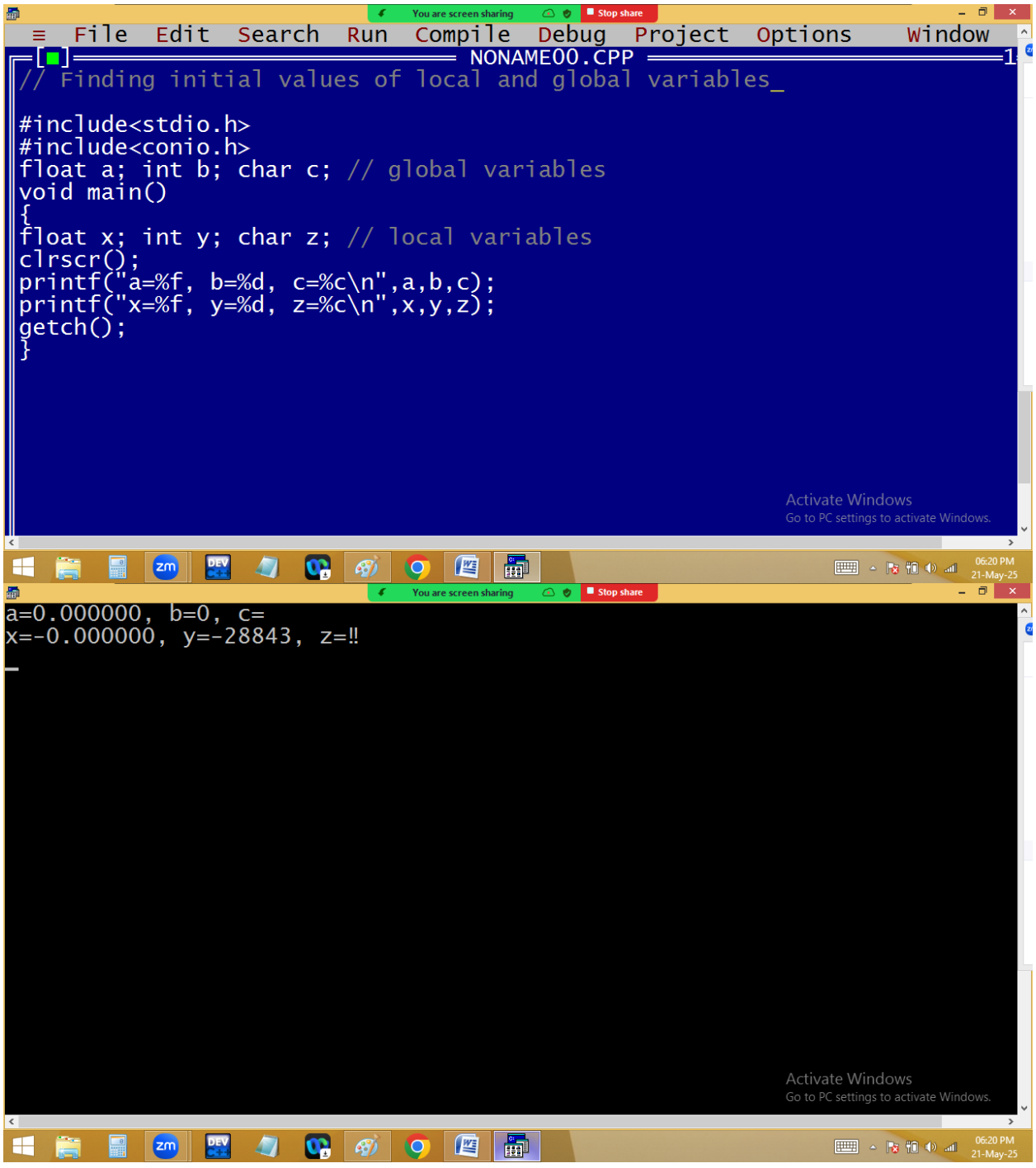
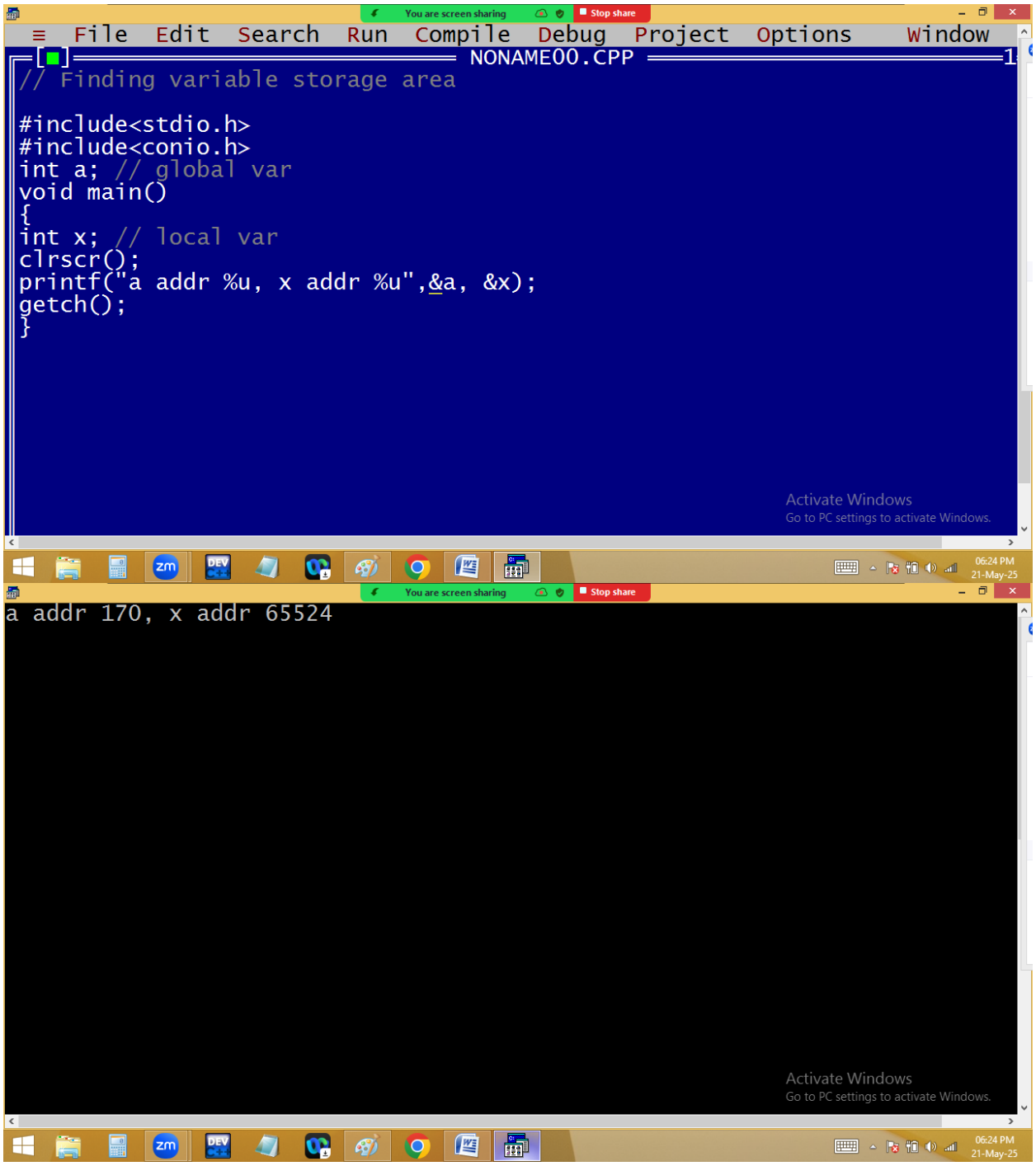
Basically the variables are divided into 2 types.

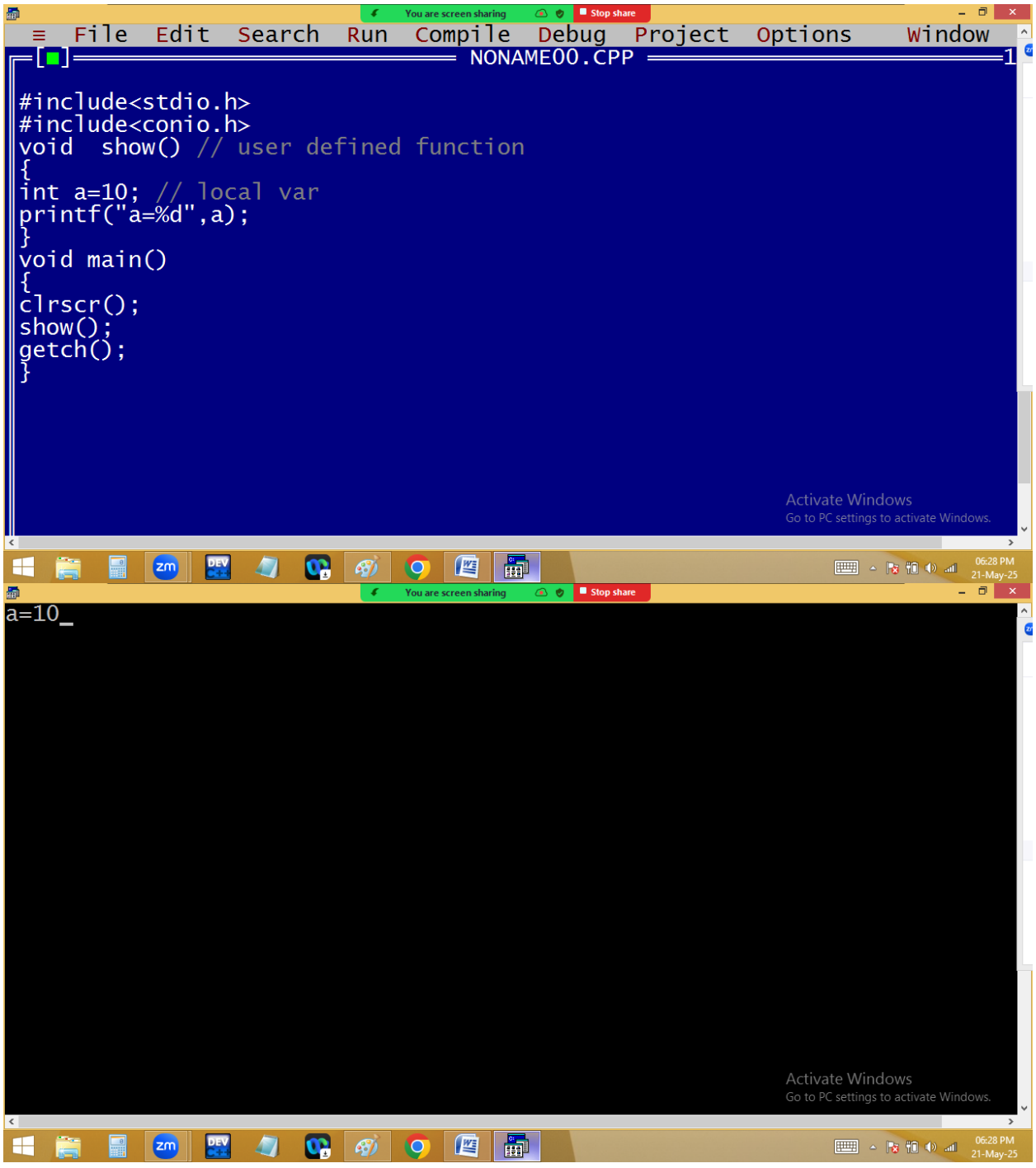
1. Local variables
2. Global variables

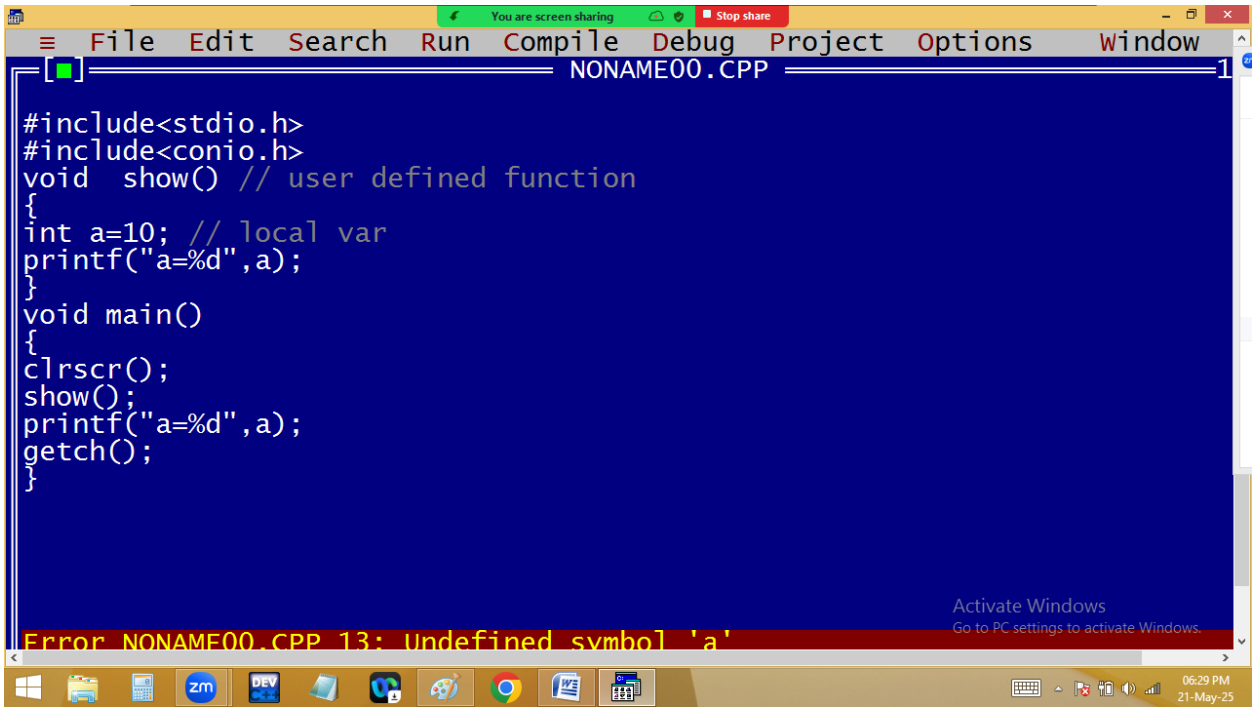
	Local variables	Global variables
Declaration	**Within function / { }**	**Top of the program**
Storage area	**Stack area**	**Data area**
Initial values [starting values]	**Garbage values**	**Int-0, float-0.000000, Char - blank**
Scope-where we can access	**Within function only**	**anywhere**
Life-time – when the variable deleted from memory	**After function execution**	**After main( ) execution**


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NONAME00.CPP
// Finding initial values of local and global variables_
#include<stdio.h>
#include<conio.h>
float a; int b; char c; // global variables
void main()
{
float x; int y; char z; // local variables
clrscr();
printf("a=%f, b=%d, c=%c\n",a,b,c);
printf("x=%f, y=%d, z=%c\n",x,y,z);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
06:20 PM
21-May-25
You are screen sharing
Stop share
a=0.000000, b=0, c=
x=-0.000000, y=-28843, z=‼
Activate Windows
Go to PC settings to activate Windows.
06:20 PM
21-May-25

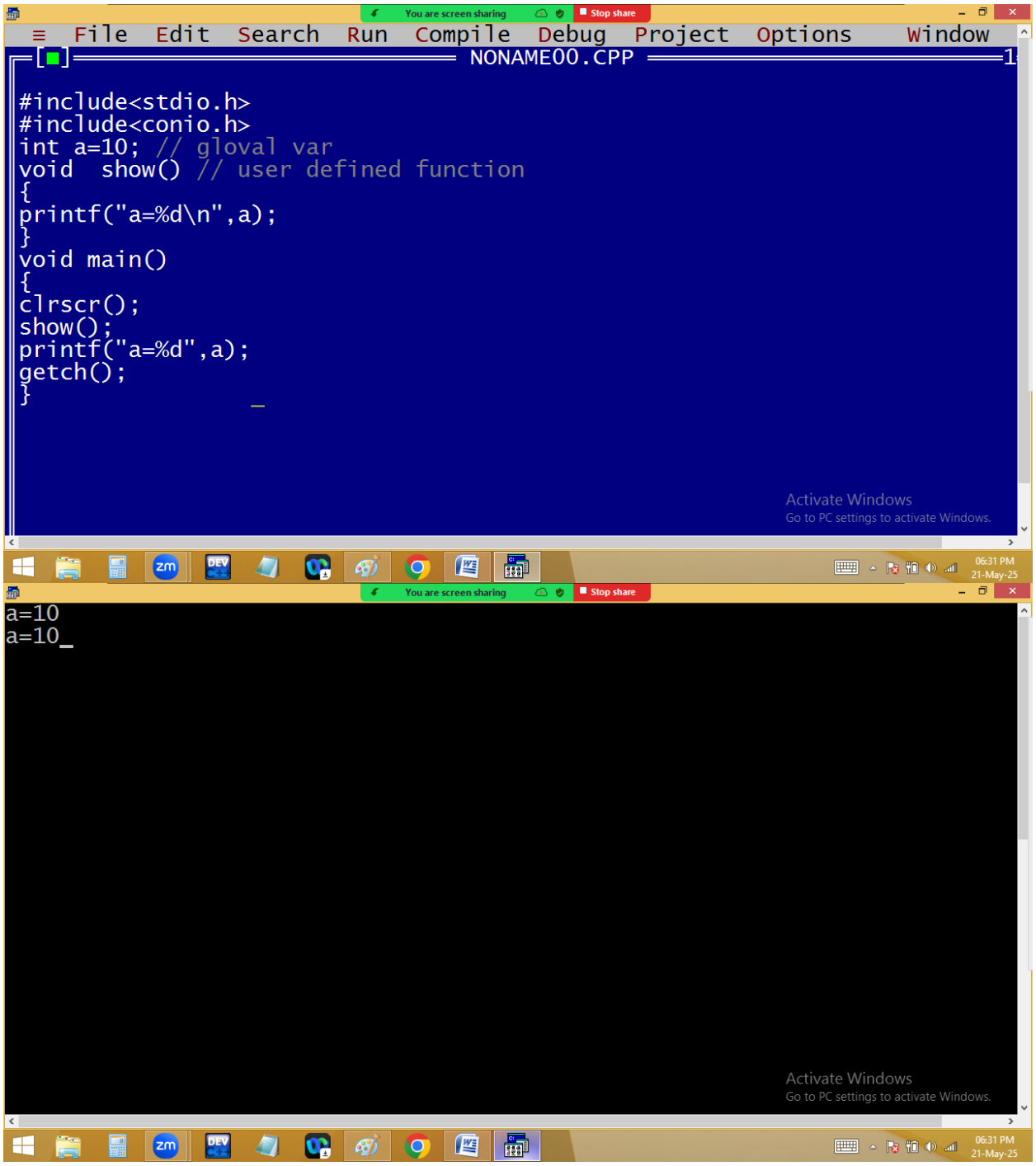


Finding scope of a local variable:

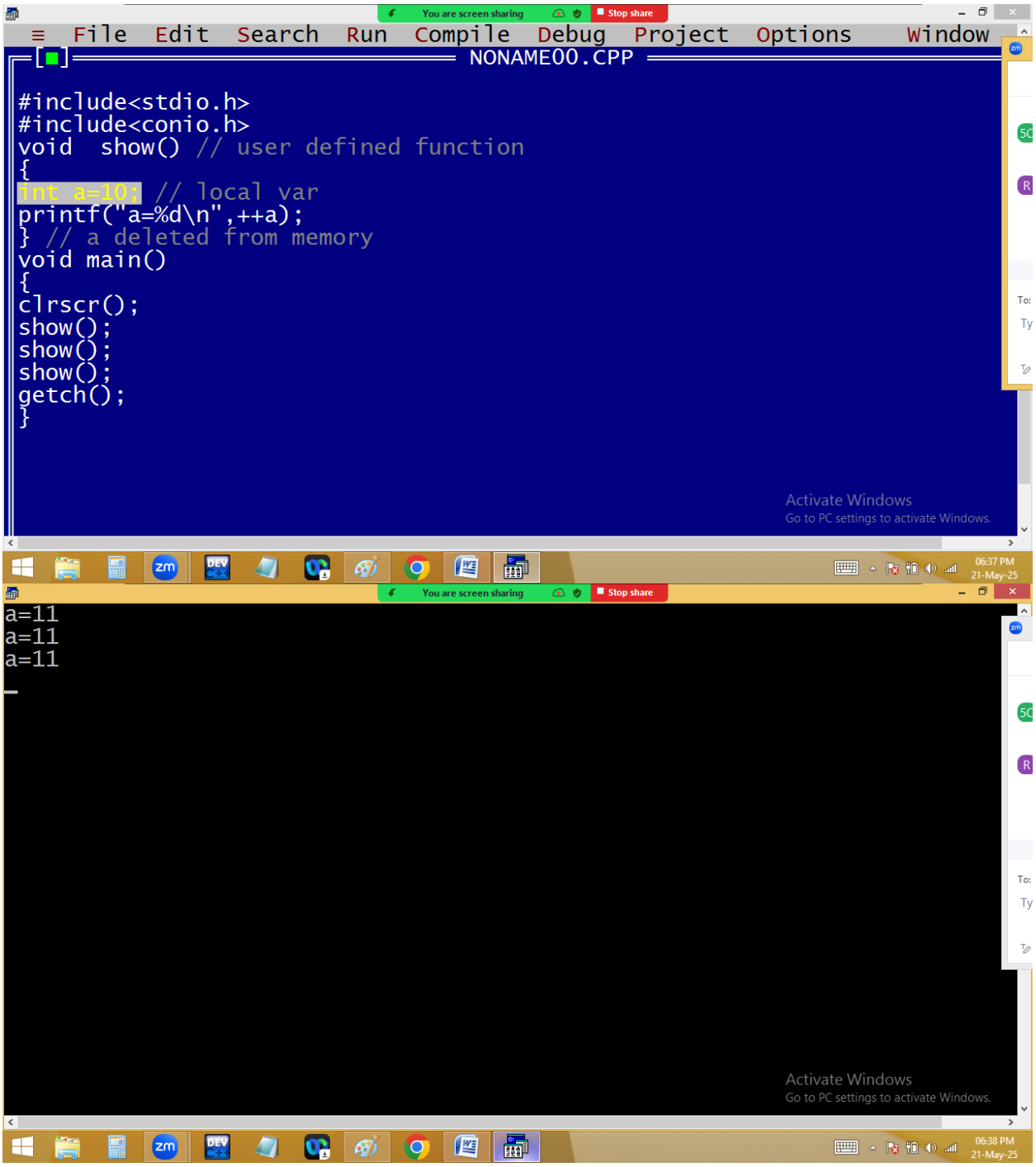

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NONAME00.CPP
#include<stdio.h>
#include<conio.h>
void show() // user defined function
{
int a=10; // local var
printf("a=%d",a);
}
void main()
{
clrscr();
show();
getch();
}
Activate Windows
Go to PC settings to activate Windows.
06:28 PM
21-May-25
You are screen sharing
Stop share
a=10
Activate Windows
Go to PC settings to activate Windows.
06:28 PM
21-May-25

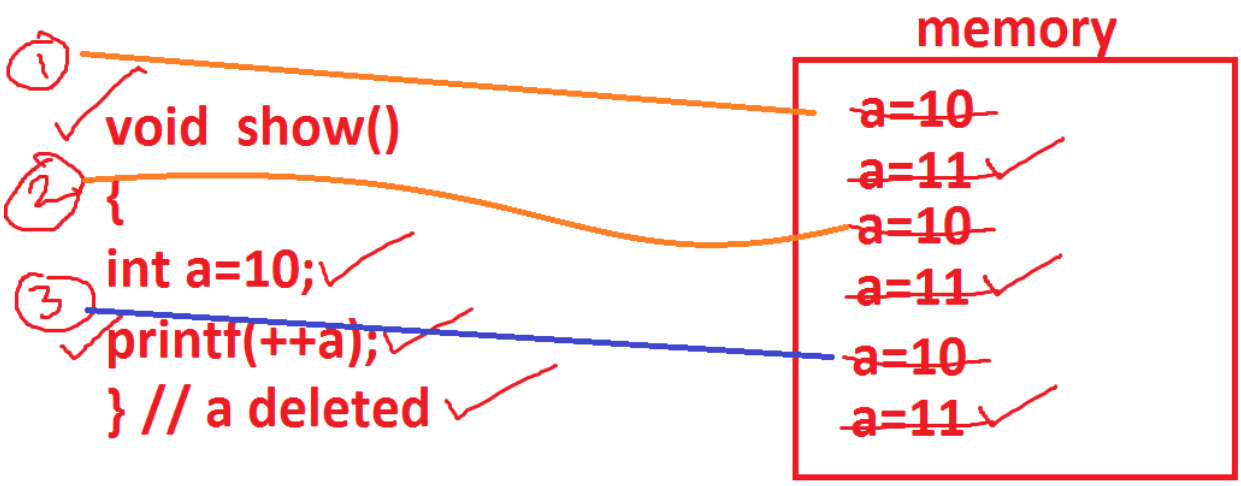


Finding scope of global variable:

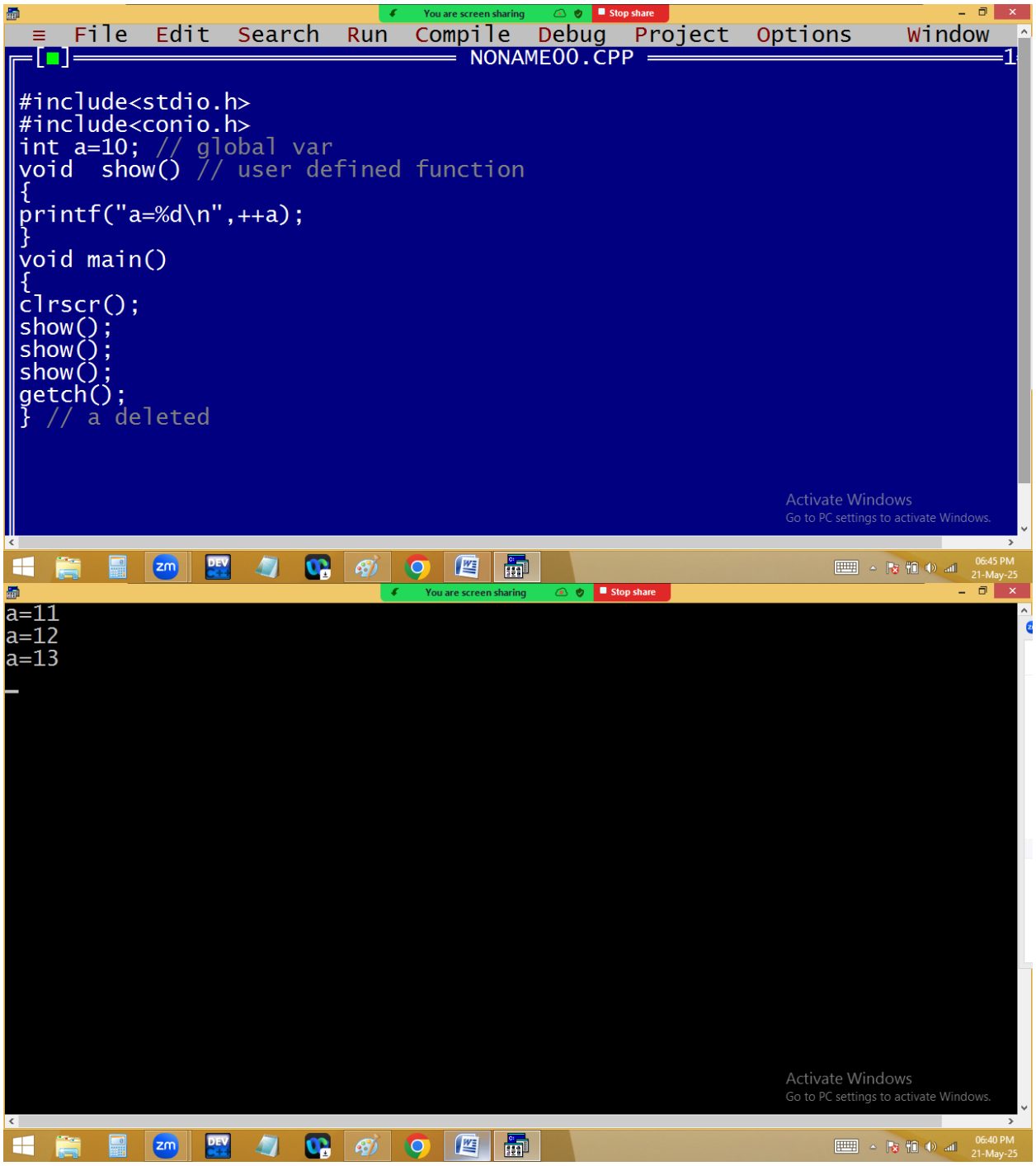


Finding life time of a local variable:

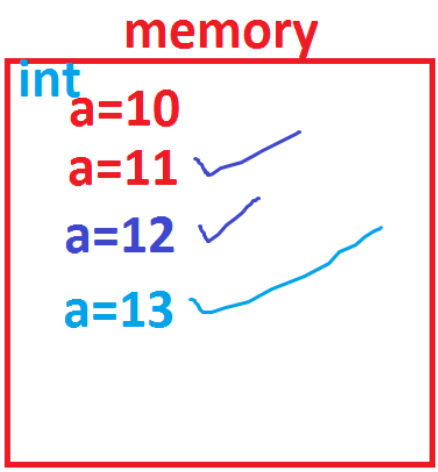

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Stop share
File Edit Search Run Compile Debug Project Options Window
NONAME00.CPP
#include<stdio.h>
#include<conio.h>
void show() // user defined function
{
int a=10; // local var
printf("a=%d\n",++a);
} // a deleted from memory
void main()
{
clrscr();
show();
show();
show();
getch();
}
Activate Windows
Go to PC settings to activate Windows.
06:37 PM
21-May-25
You are screen sharing
Stop share
a=11
a=11
a=11
Activate Windows
Go to PC settings to activate Windows.
06:38 PM
21-May-25



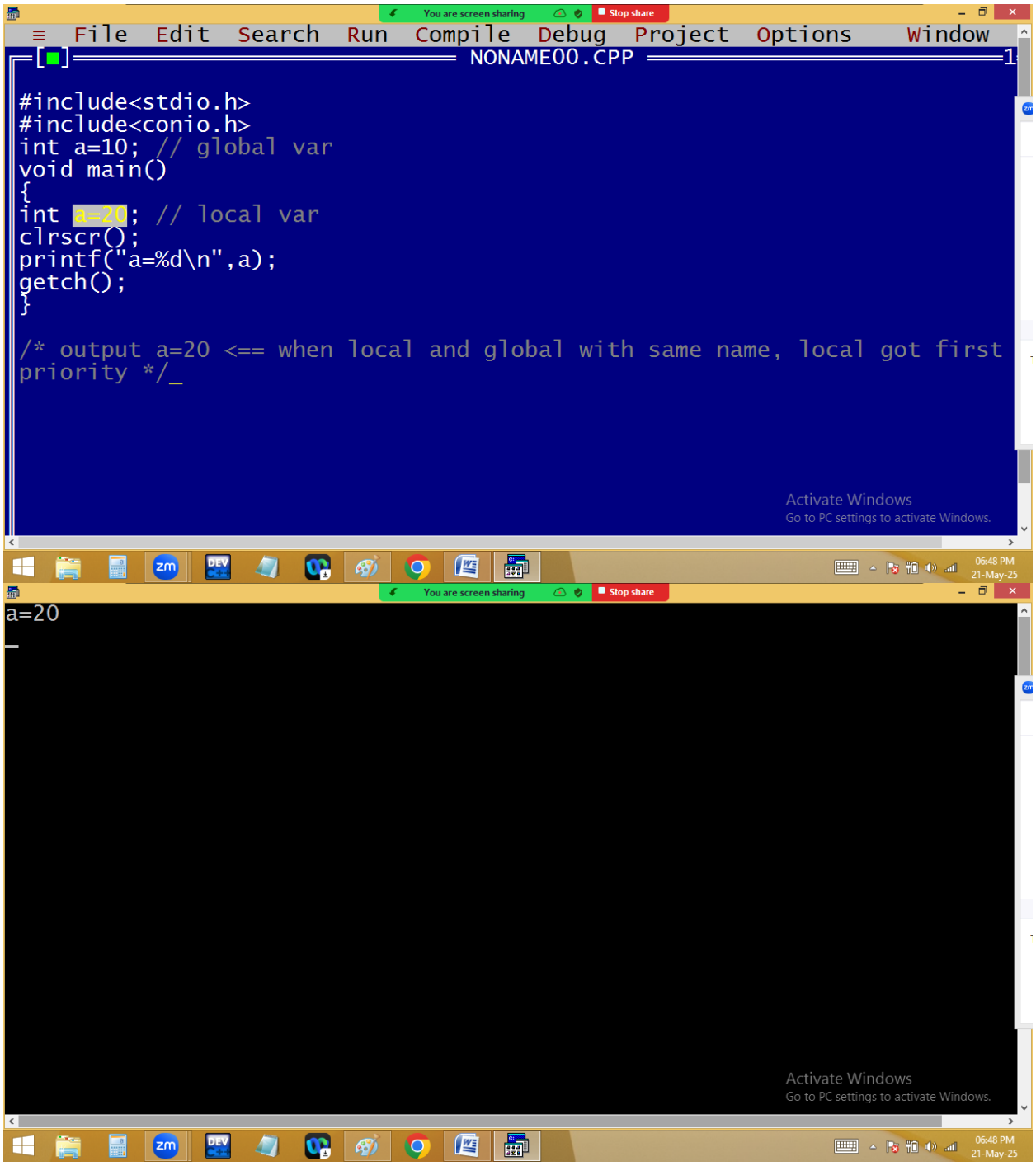
Finding life time of a global variable:

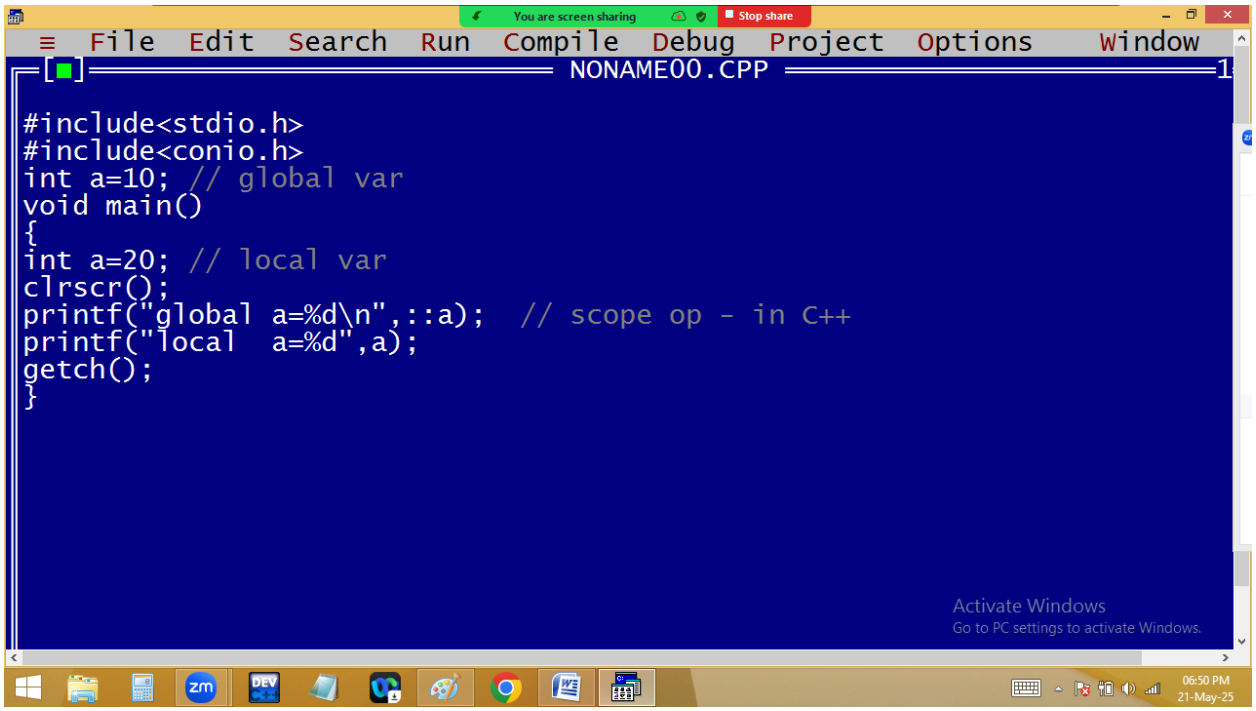

You are screen sharing
Stop share
File Edit Search Run Compile Debug Project Options Window
NONAME00.CPP
#include<stdio.h>
#include<conio.h>
int a=10; // global var
void show() // user defined function
{
printf("a=%d\n",++a);
}
void main()
{
clrscr();
show();
show();
show();
getch();
} // a deleted
Activate Windows
Go to PC settings to activate Windows.
06:45 PM
21-May-25
You are screen sharing
Stop share
a=11
a=12
a=13
Activate Windows
Go to PC settings to activate Windows.
06:40 PM
21-May-25

```
int a=10; global
void  show()
{
printf(++a);
}
```



when main() closed a deleted

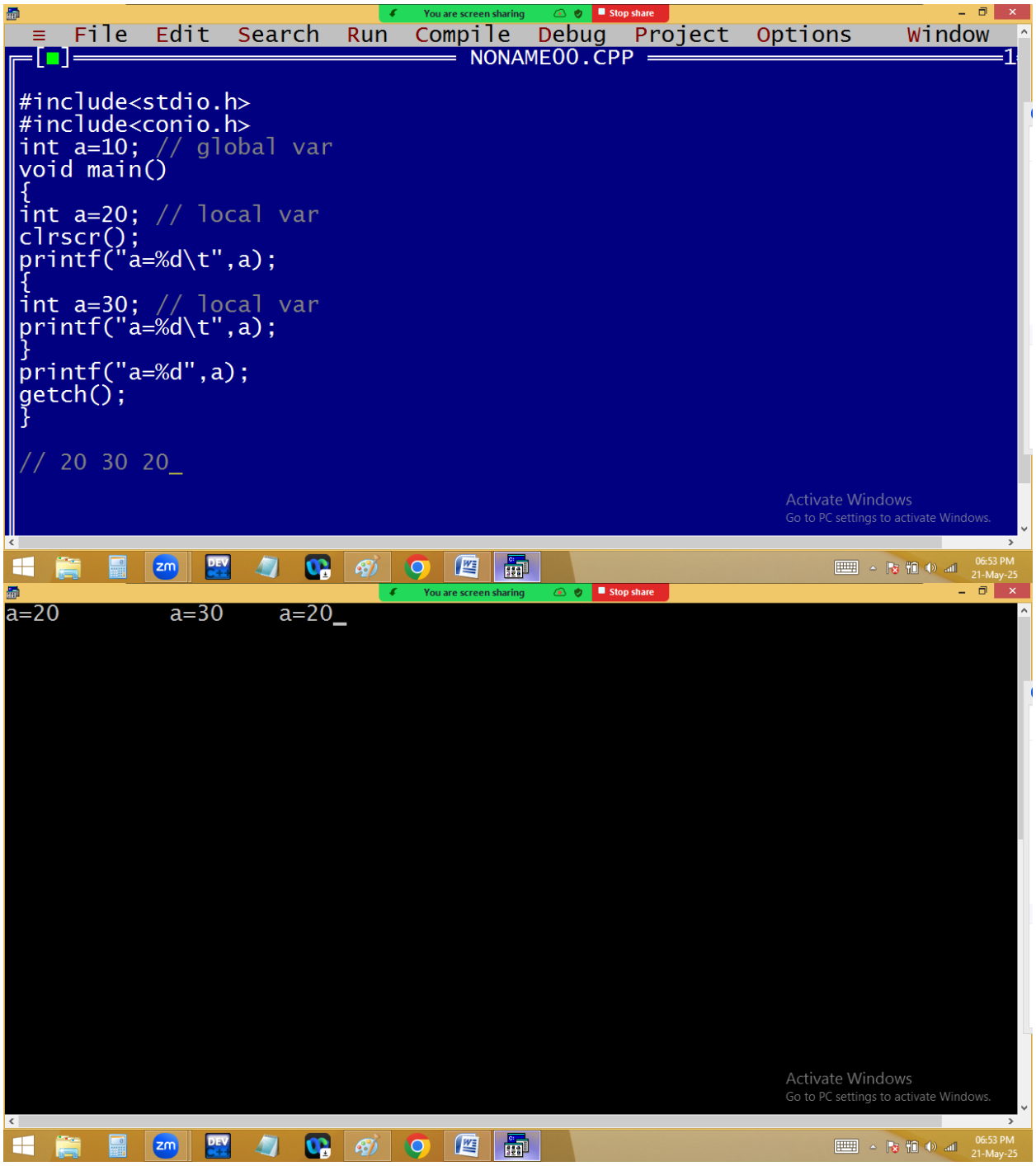

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File Edit Search Run Compile Debug Project Options Window
NONAME00.CPP
#include<stdio.h>
#include<conio.h>
int a=10; // global var
void main()
{
int a=20; // local var
clrscr();
printf("a=%d\n",a);
getch();
}
/* output a=20 <== when local and global with same name, local got first priority */
Activate Windows
Go to PC settings to activate Windows.
06:48 PM
21-May-25
You are screen sharing
Stop share
a=20
Activate Windows
Go to PC settings to activate Windows.
06:48 PM
21-May-25



output:

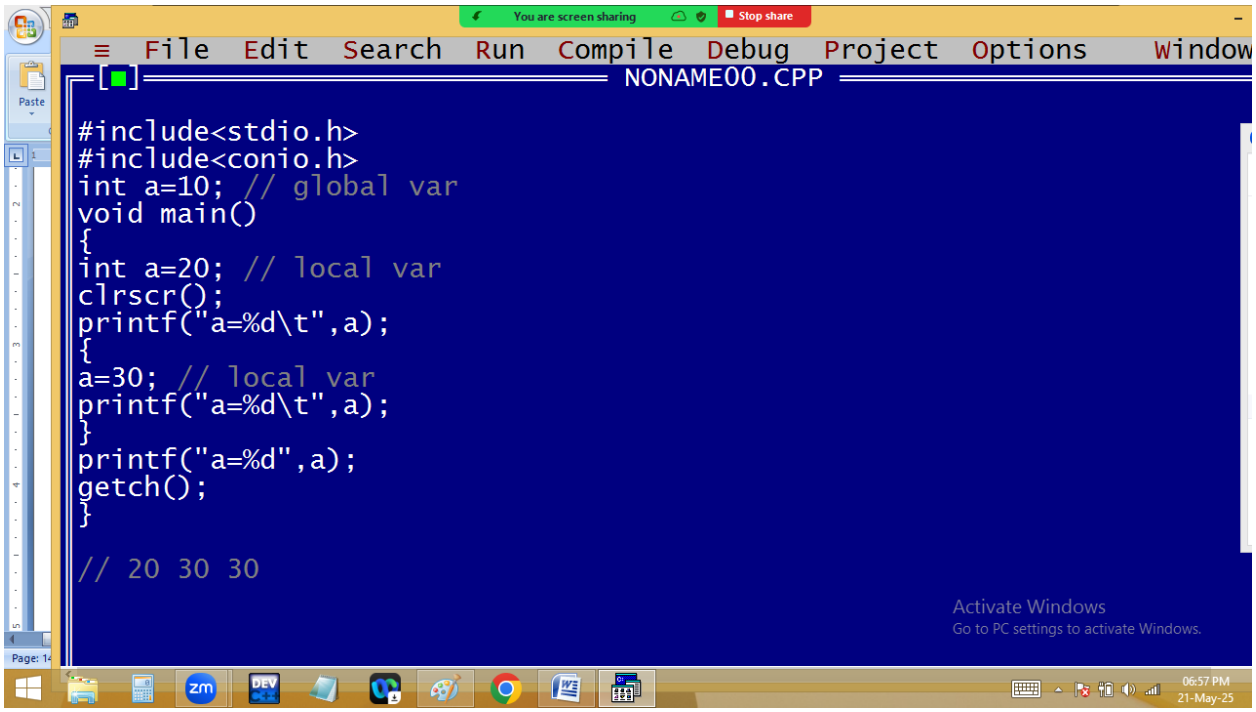
Global a=10

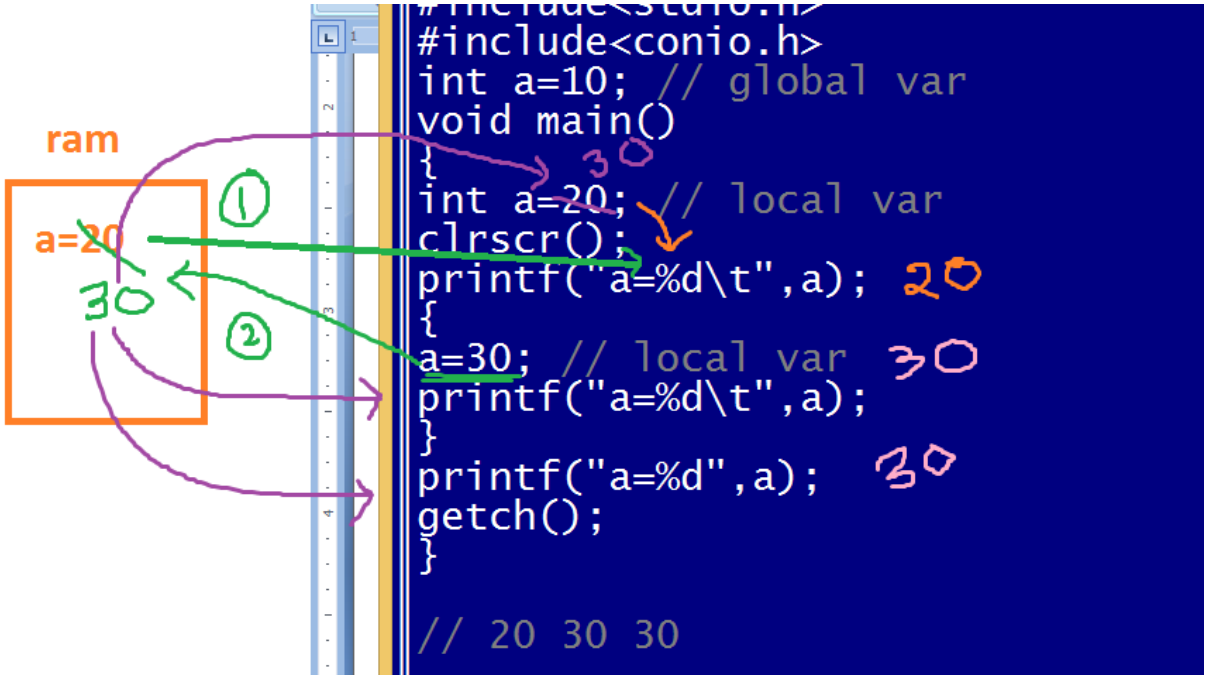
Local a=20

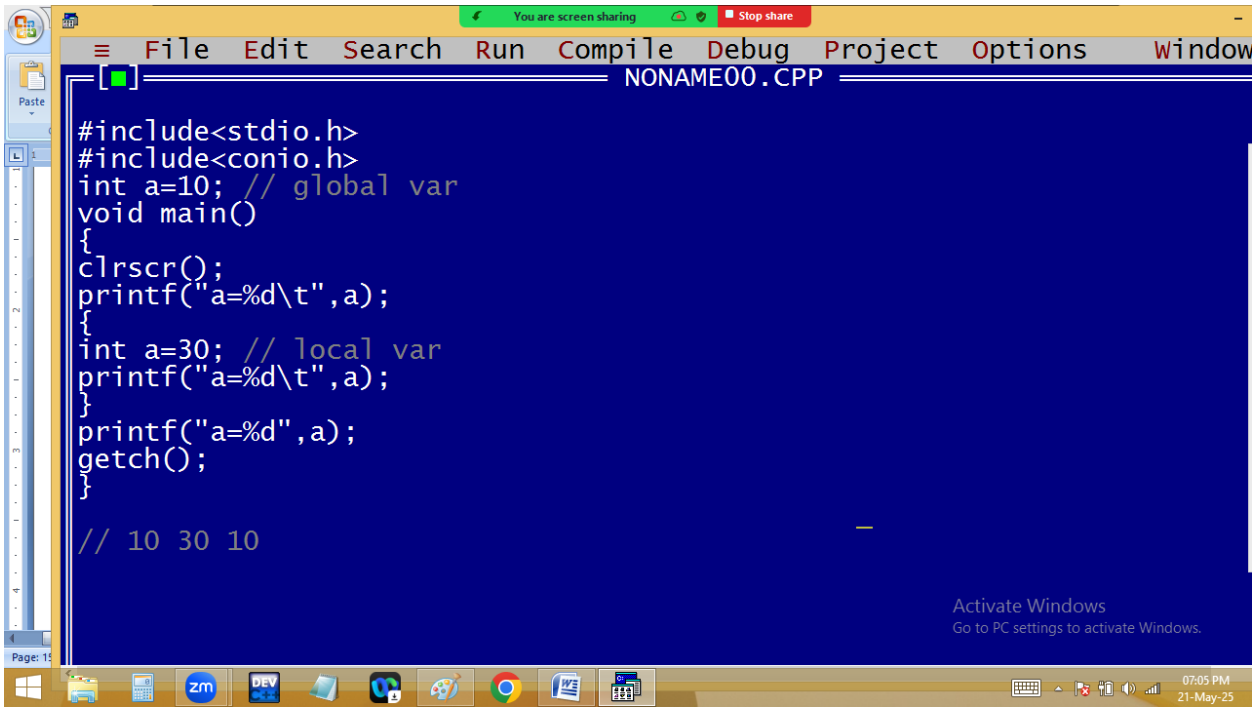

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File Edit Search Run Compile Debug Project Options Window
NONAME00.CPP
#include<stdio.h>
#include<conio.h>
int a=10; // global var
void main()
{
int a=20; // local var
clrscr();
printf("a=%d\t",a);
{
int a=30; // local var
printf("a=%d\t",a);
}
printf("a=%d",a);
getch();
}
// 20 30 20_
Activate Windows
Go to PC settings to activate Windows.
06:53 PM
21-May-25
You are screen sharing
Stop share
a=20 a=30 a=20_
Activate Windows
Go to PC settings to activate Windows.
06:53 PM
21-May-25

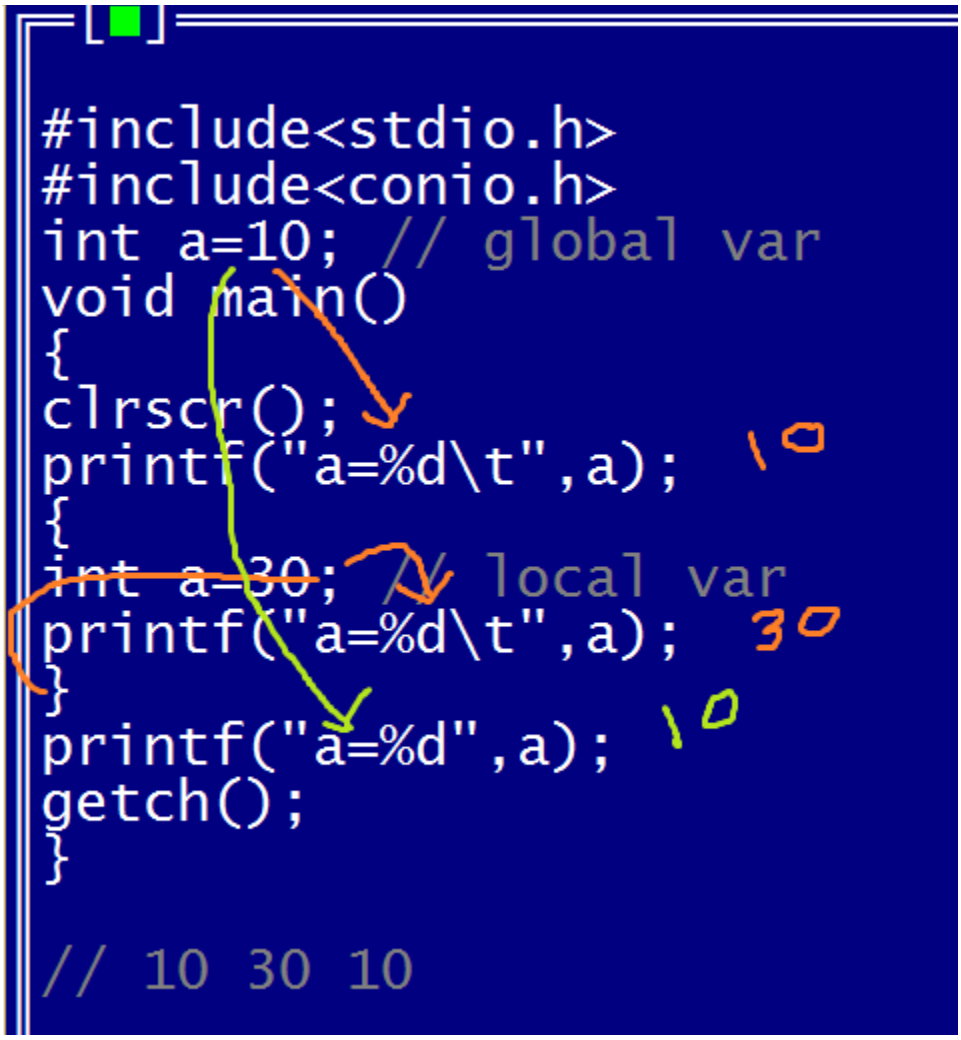
```c
#include<stdio.h>
#include<conio.h>
int a=10; // global var
void main()
{
int a=20; // local var
clrscr();
printf("a=%d\t",a);
{
int a=30; // local var
printf("a=%d\t",a);
}
printf("a=%d",a);
getch();
}
```

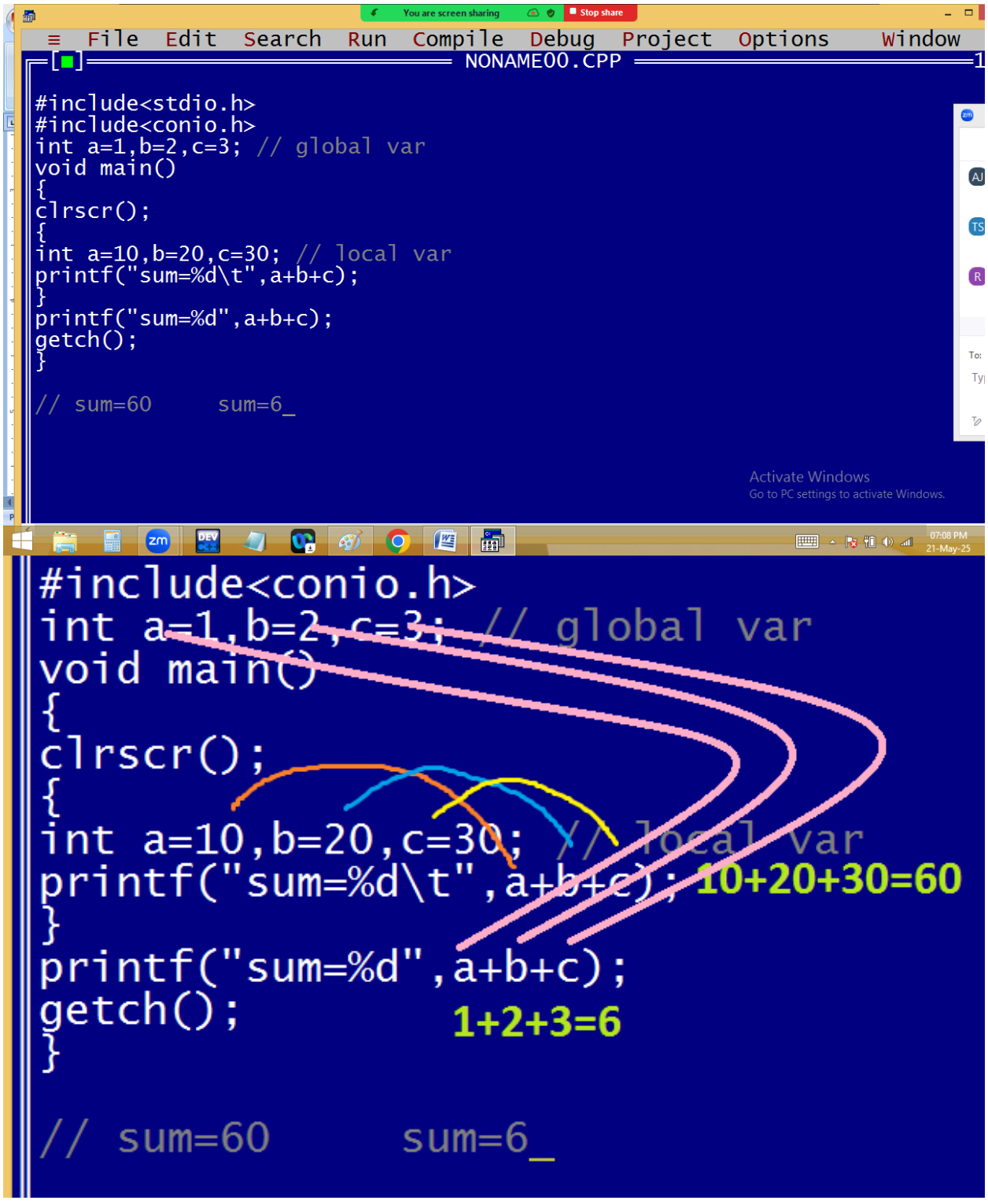
20 30 20

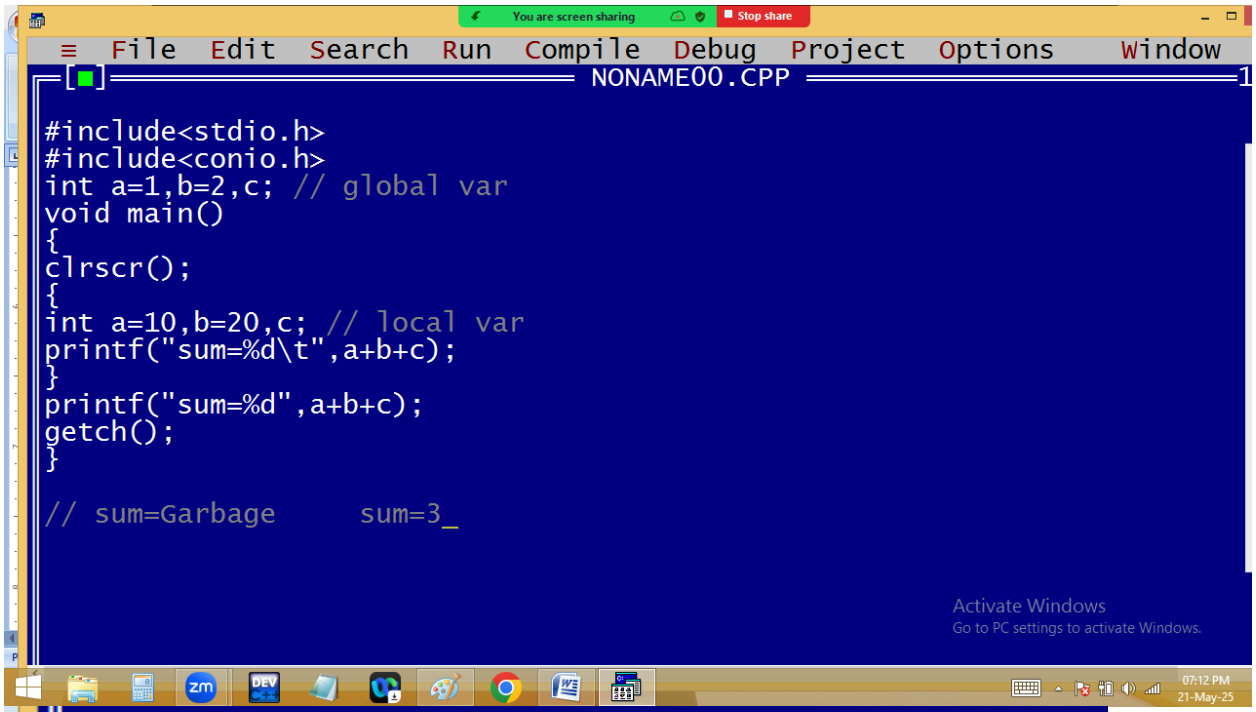


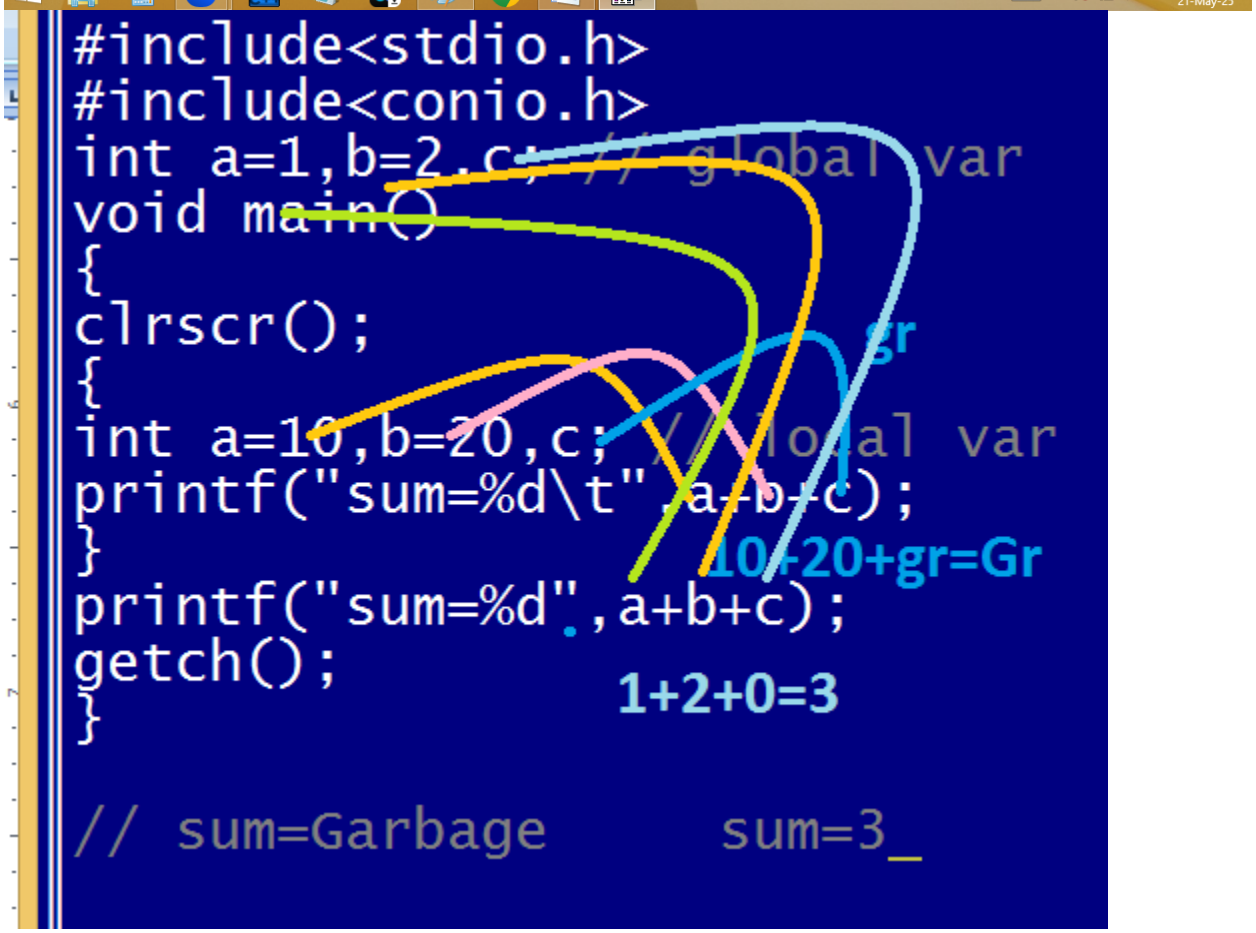

ram
a=20
30
1
2
#include<conio.h>
int a=10; // global var
void main()
{
int a=20; // local var
clrscr();
printf("a=%d\t",a);
{
a=30; // local var
printf("a=%d\t",a);
}
printf("a=%d",a);
getch();
}
// 20 30 30

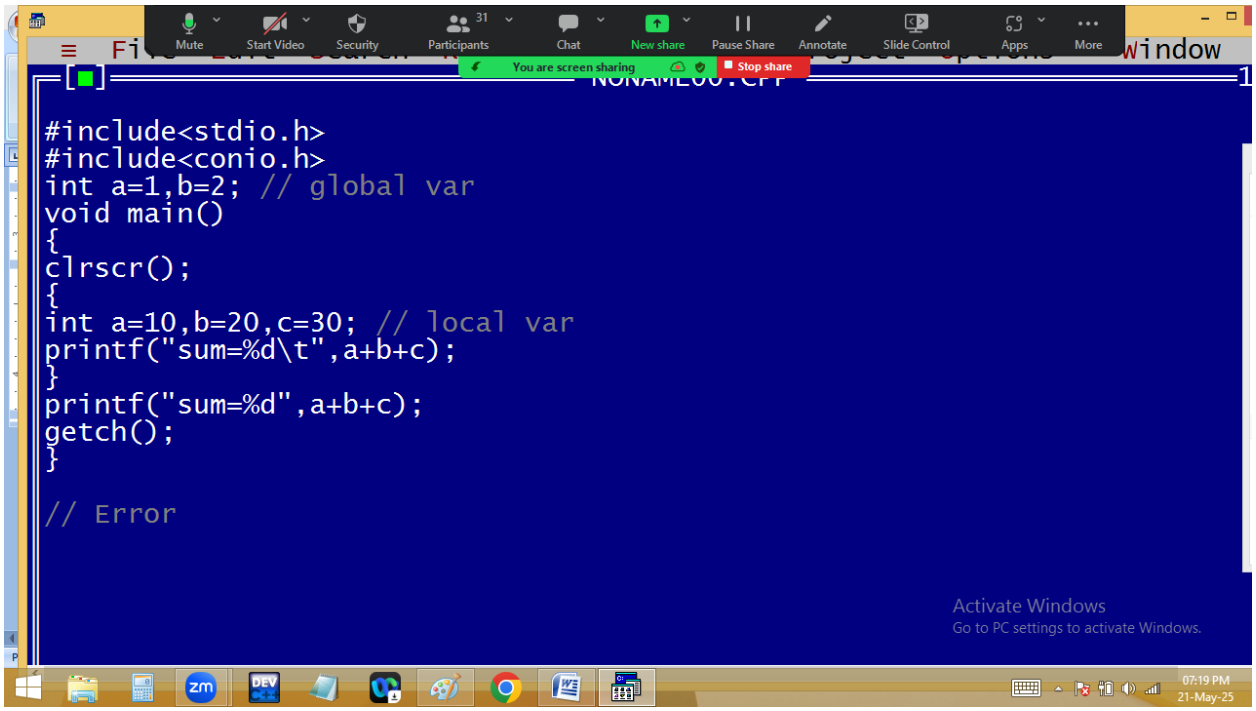

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File Edit Search Run Compile Debug Project Options Window
NONAME00.CPP
#include<stdio.h>
#include<conio.h>
int a=10; // global var
void main()
{
clrscr();
printf("a=%d\t",a);
{
int a=30; // local var
printf("a=%d\t",a);
}
printf("a=%d",a);
getch();
}
// 10 30 10
Activate Windows
Go to PC settings to activate Windows.
07:05 PM
21-May-25


#include<stdio.h>
#include<conio.h>
int a=10; // global var
void main()
{
clrscr();
printf("a=%d\t",a);
{
int a=30; // local var
printf("a=%d\t",a);
}
printf("a=%d",a);
getch();
}
// 10 30 10
10
30
10

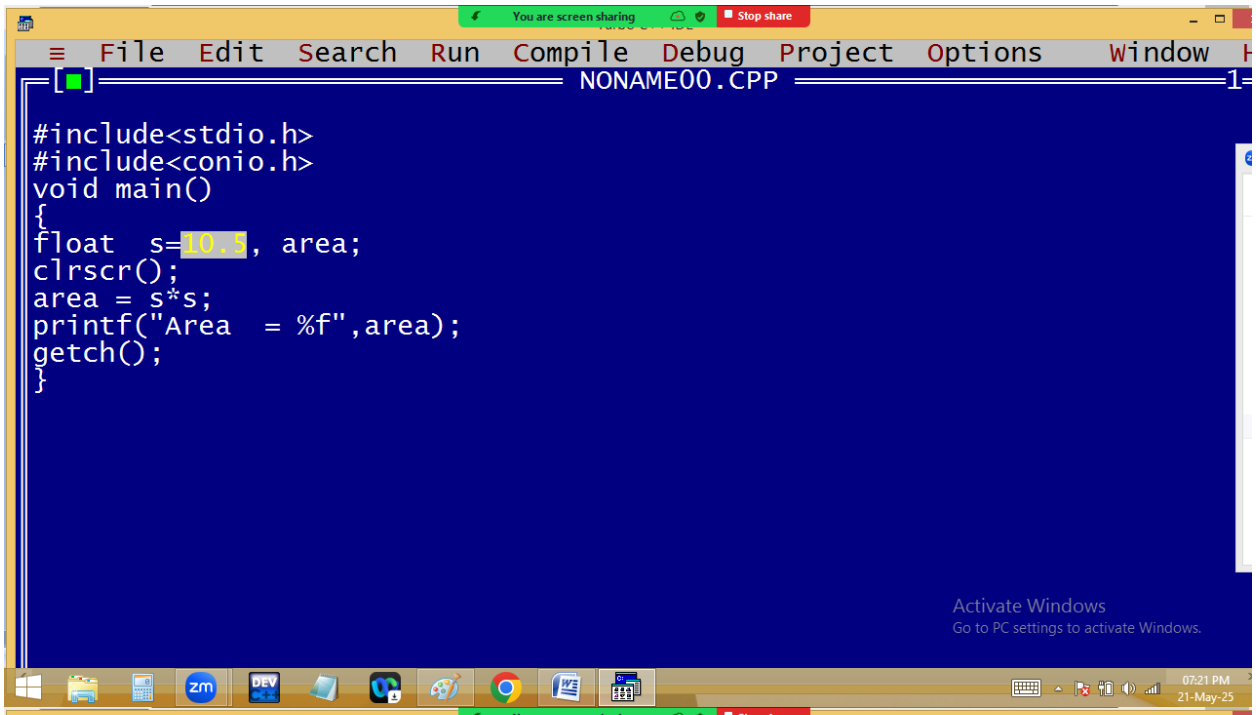

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NONAME00.CPP
#include<stdio.h>
#include<conio.h>
int a=1,b=2,c=3; // global var
void main()
{
clrscr();
{
int a=10,b=20,c=30; // local var
printf("sum=%d\t",a+b+c);
}
printf("sum=%d",a+b+c);
getch();
}
// sum=60 sum=6_
Activate Windows
Go to PC settings to activate Windows.
#include<conio.h>
int a=1,b=2,c=3; // global var
void main()
{
clrscr();
{
int a=10,b=20,c=30; // local var
printf("sum=%d\t",a+b+c); 10+20+30=60
}
printf("sum=%d",a+b+c);
getch(); 1+2+3=6
}
// sum=60 sum=6_

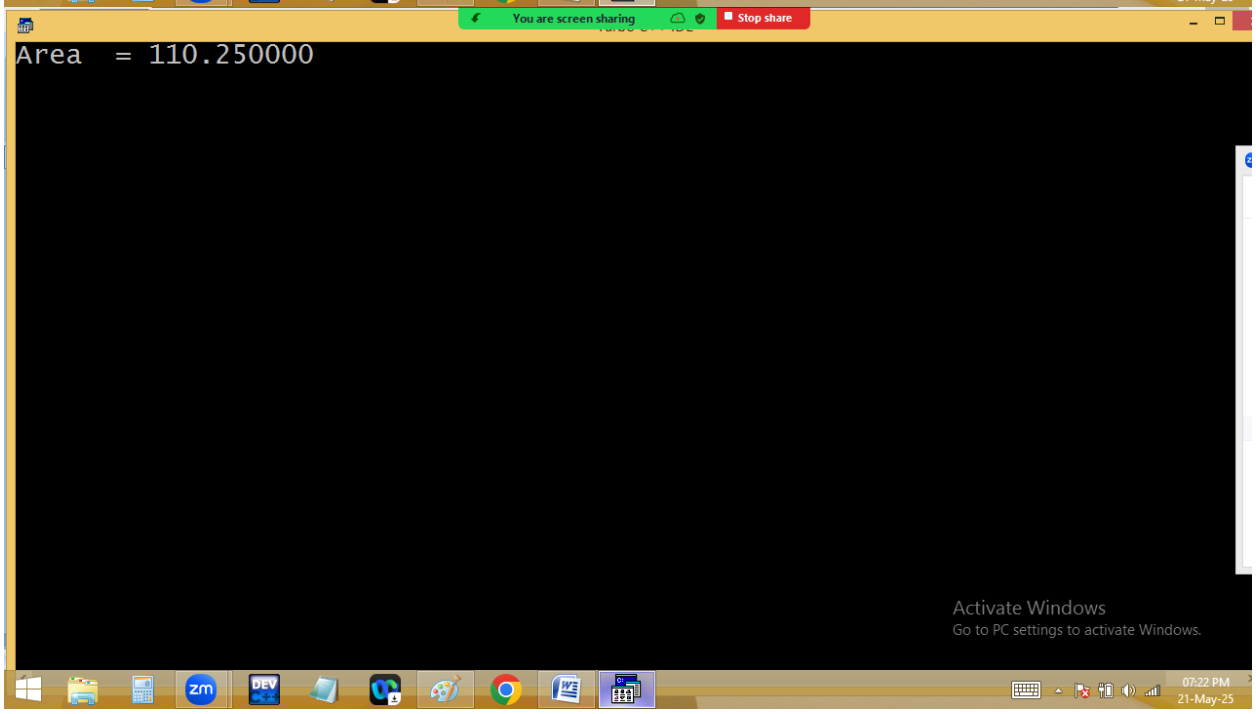

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Stop share
File Edit Search Run Compile Debug Project Options Window
NONAME00.CPP
#include<stdio.h>
#include<conio.h>
int a=1,b=2,c; // global var
void main()
{
clrscr();
{
int a=10,b=20,c; // local var
printf("sum=%d\t",a+b+c);
}
printf("sum=%d",a+b+c);
getch();
}
// sum=Garbage sum=3_
Activate Windows
Go to PC settings to activate Windows.
07:12 PM
21-May-25

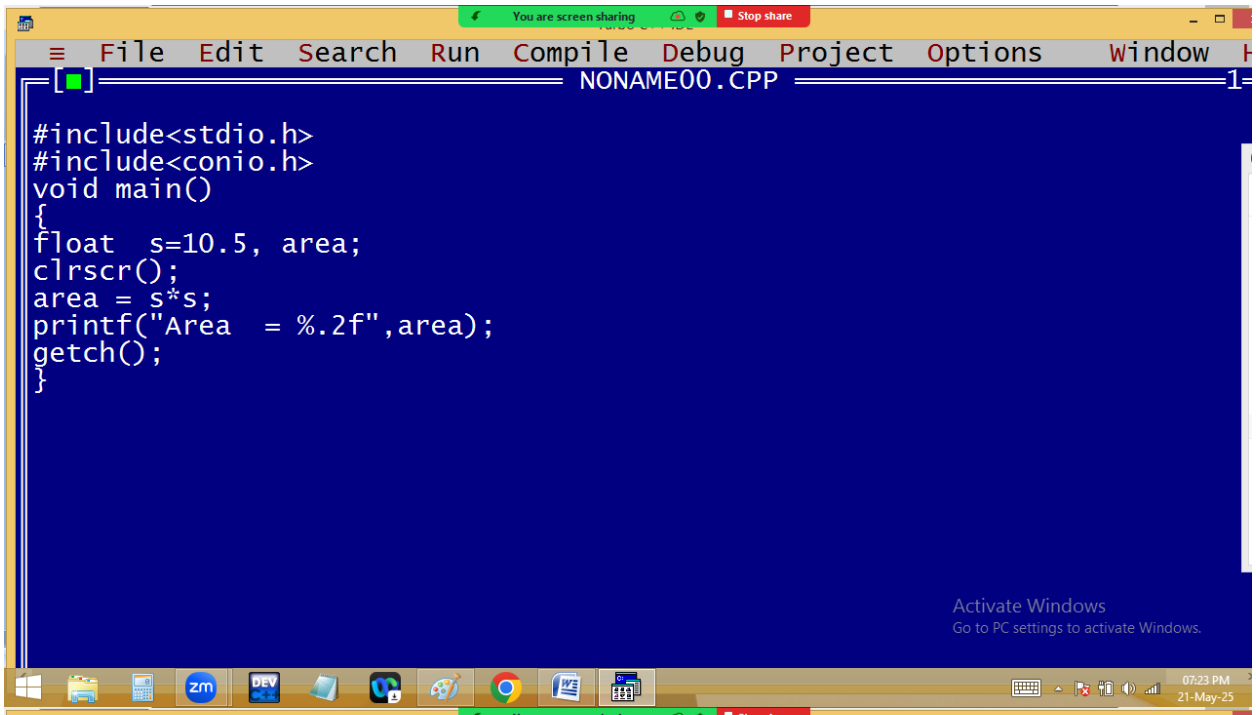

#include<stdio.h>
#include<conio.h>
int a=1,b=2,c; // global var
void main()
{
clrscr();
{
int a=10,b=20,c; // local var
printf("sum=%d\t",a+b+c);
}
printf("sum=%d",a+b+c);
getch();
}
// sum=Garbage sum=3_
gr
10+20+gr=Gr
1+2+0=3

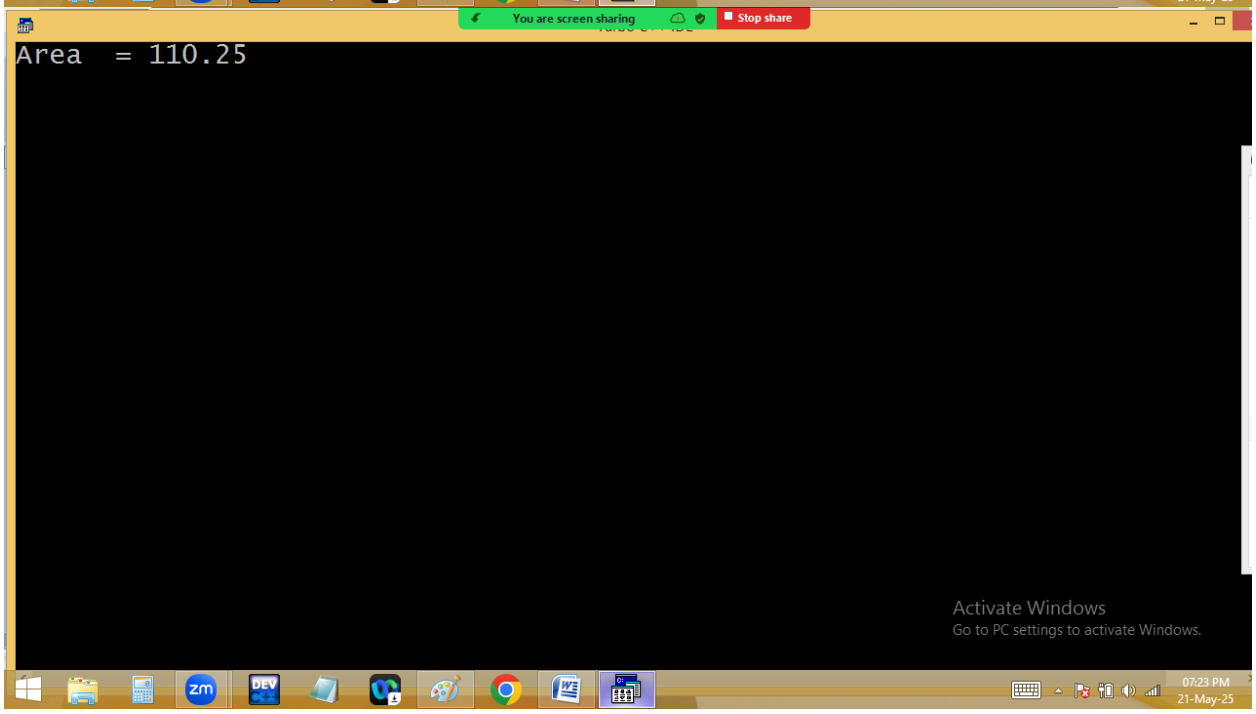


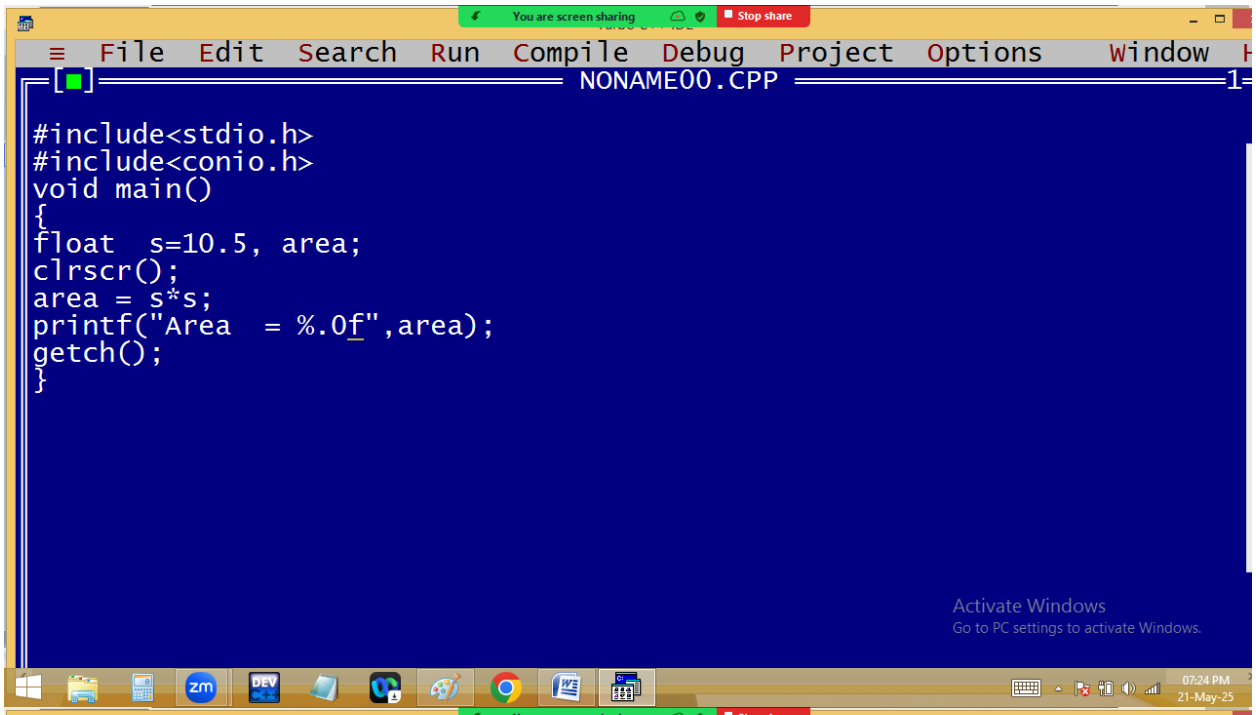
Finding area of a square?

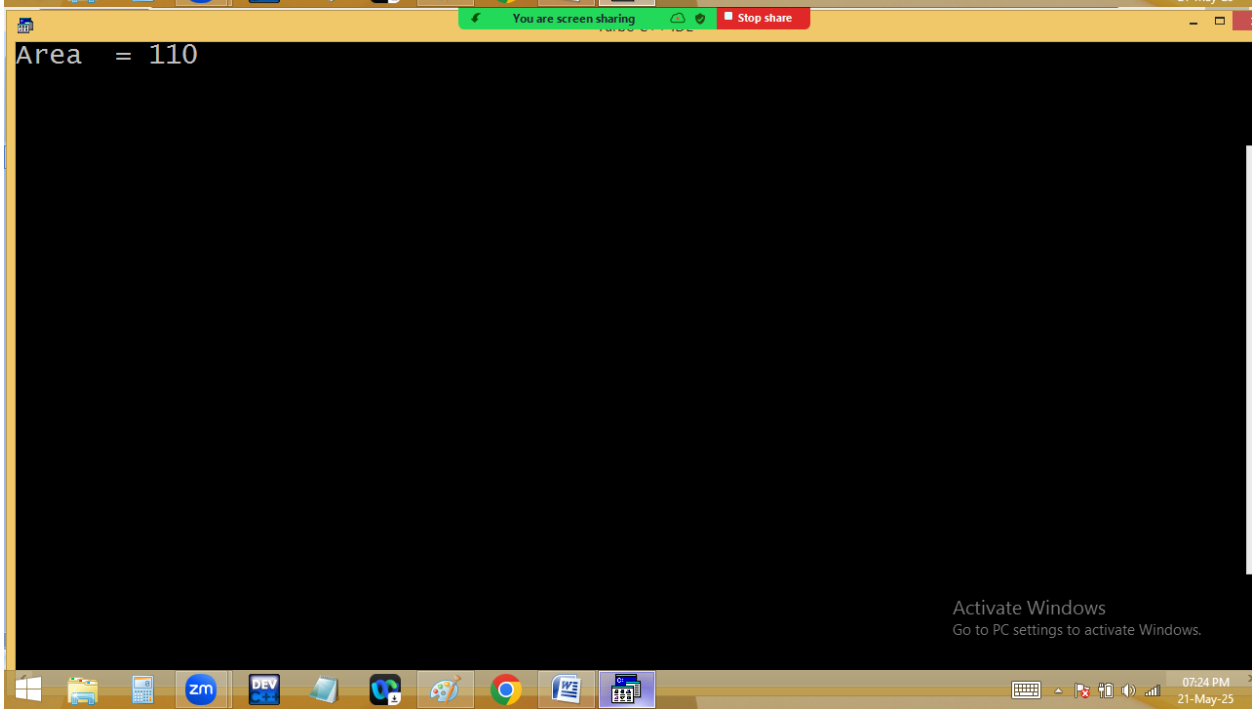

You are screen sharing
Stop share
File Edit Search Run Compile Debug Project Options Window
NONAME00.CPP
#include<stdio.h>
#include<conio.h>
void main()
{
float s=10.5, area;
clrscr();
area = s*s;
printf("Area = %f",area);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
07:21 PM
21-May-25


You are screen sharing
Stop share
Area = 110.250000
Activate Windows
Go to PC settings to activate Windows.
07:22 PM
21-May-25

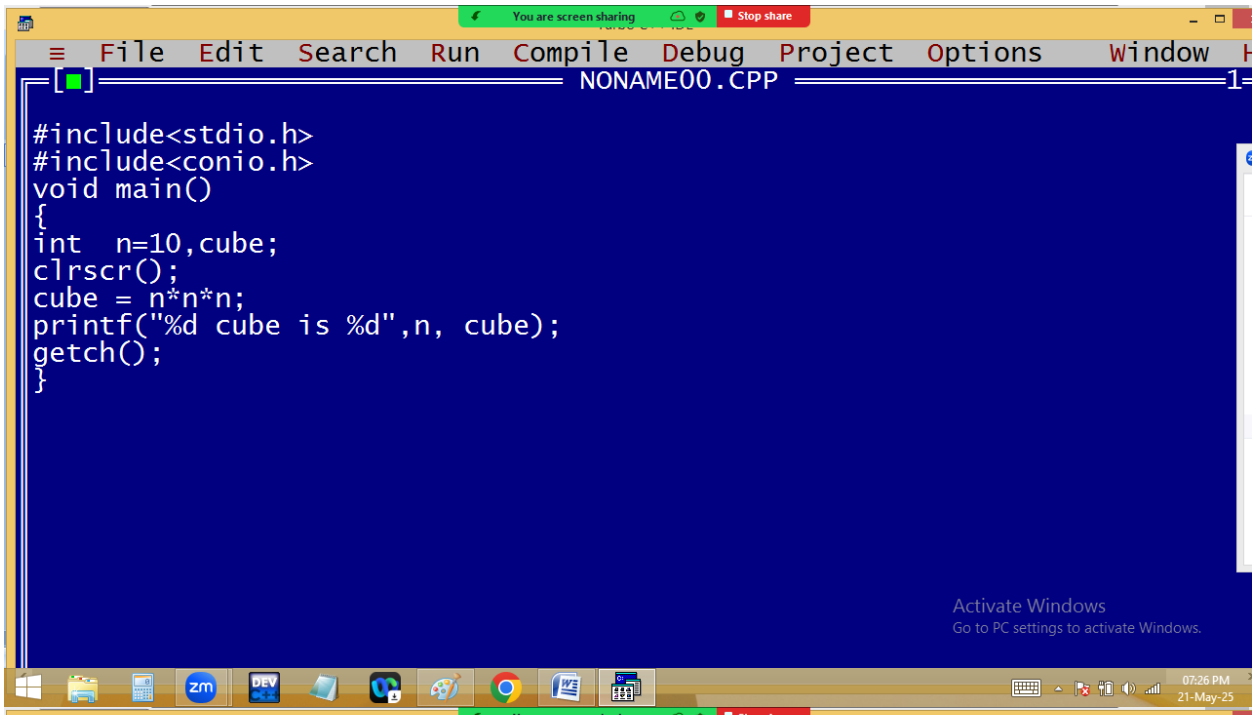

You are screen sharing
Stop share
File Edit Search Run Compile Debug Project Options Window
NONAME00.CPP
#include<stdio.h>
#include<conio.h>
void main()
{
float s=10.5, area;
clrscr();
area = s*s;
printf("Area = %.2f",area);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
07:23 PM
21-May-25

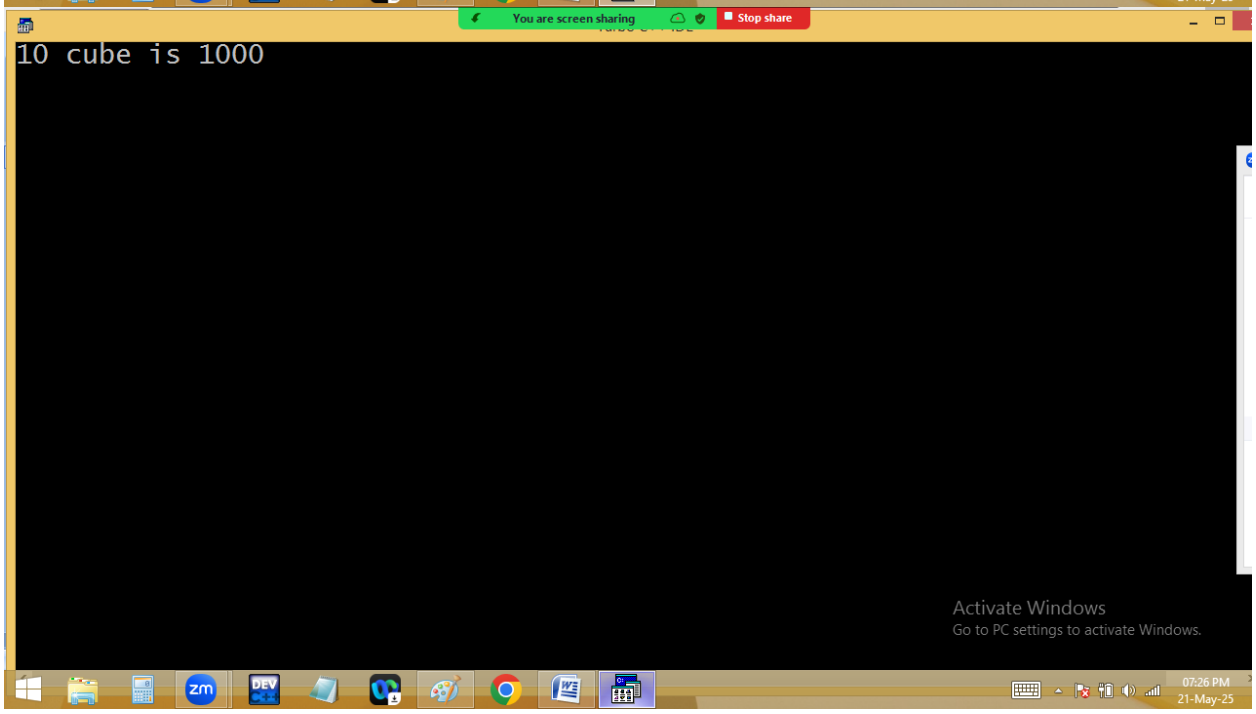

You are screen sharing
Stop share
Area = 110.25
Activate Windows
Go to PC settings to activate Windows.
07:23 PM
21-May-25

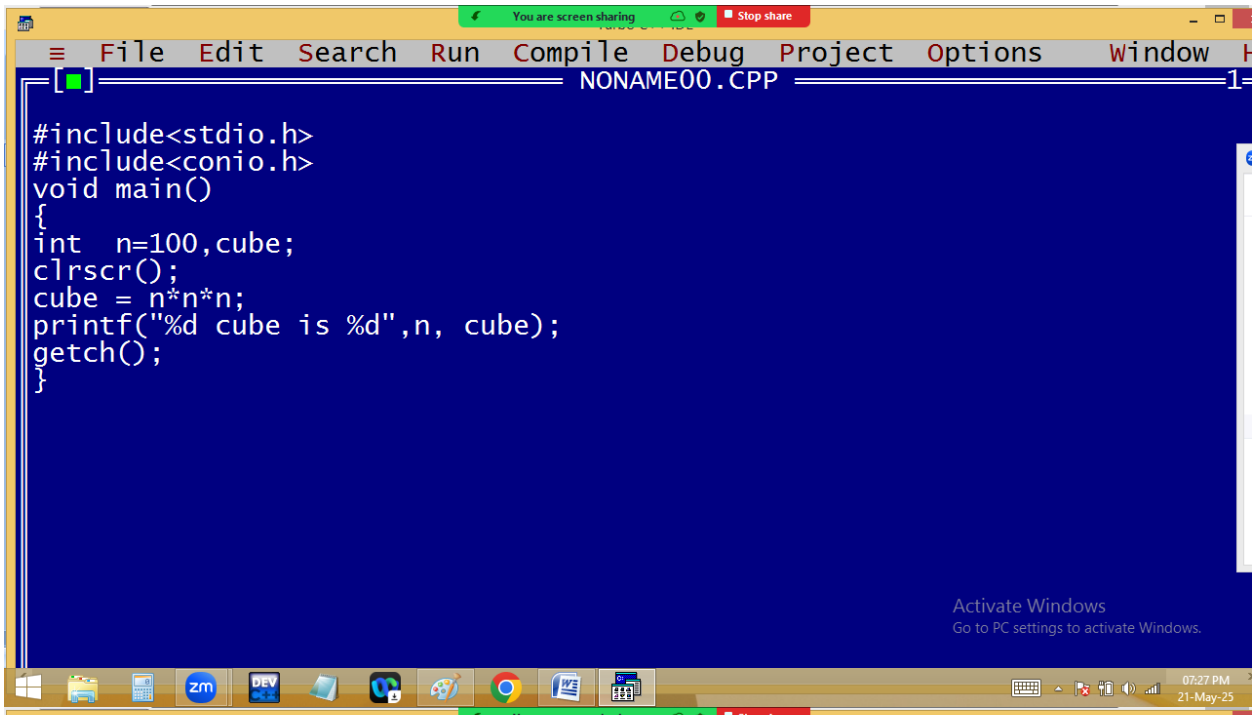


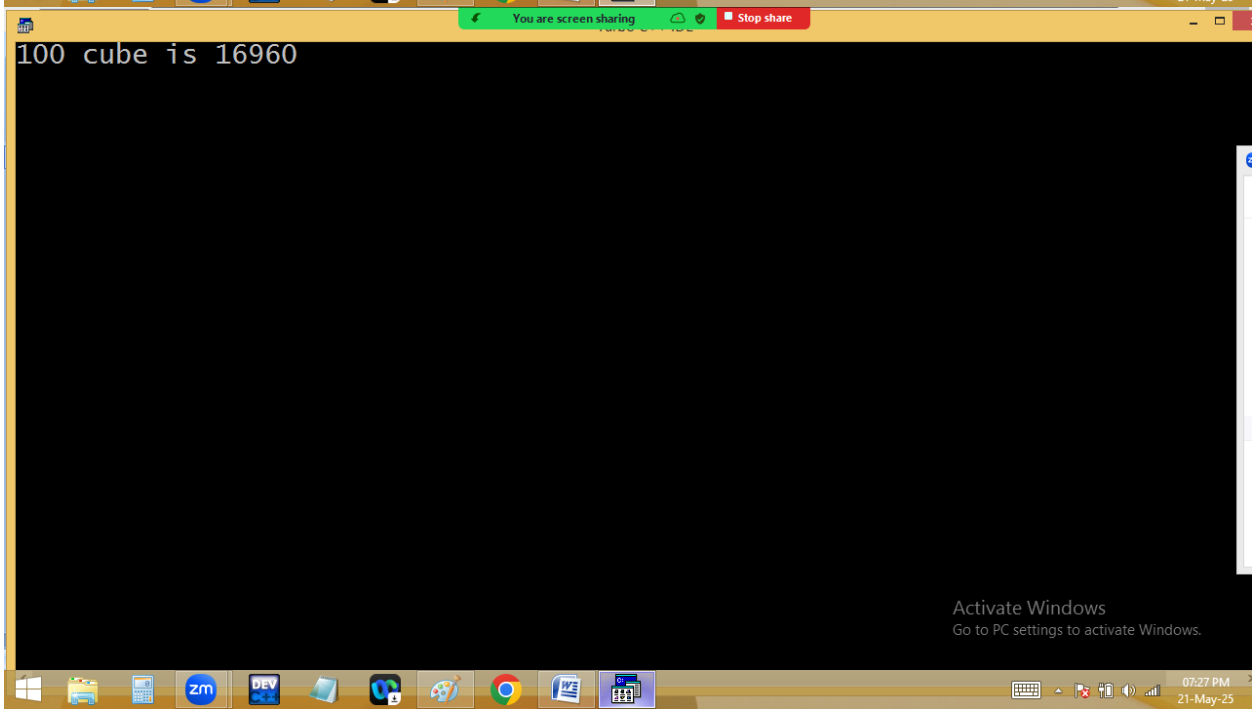


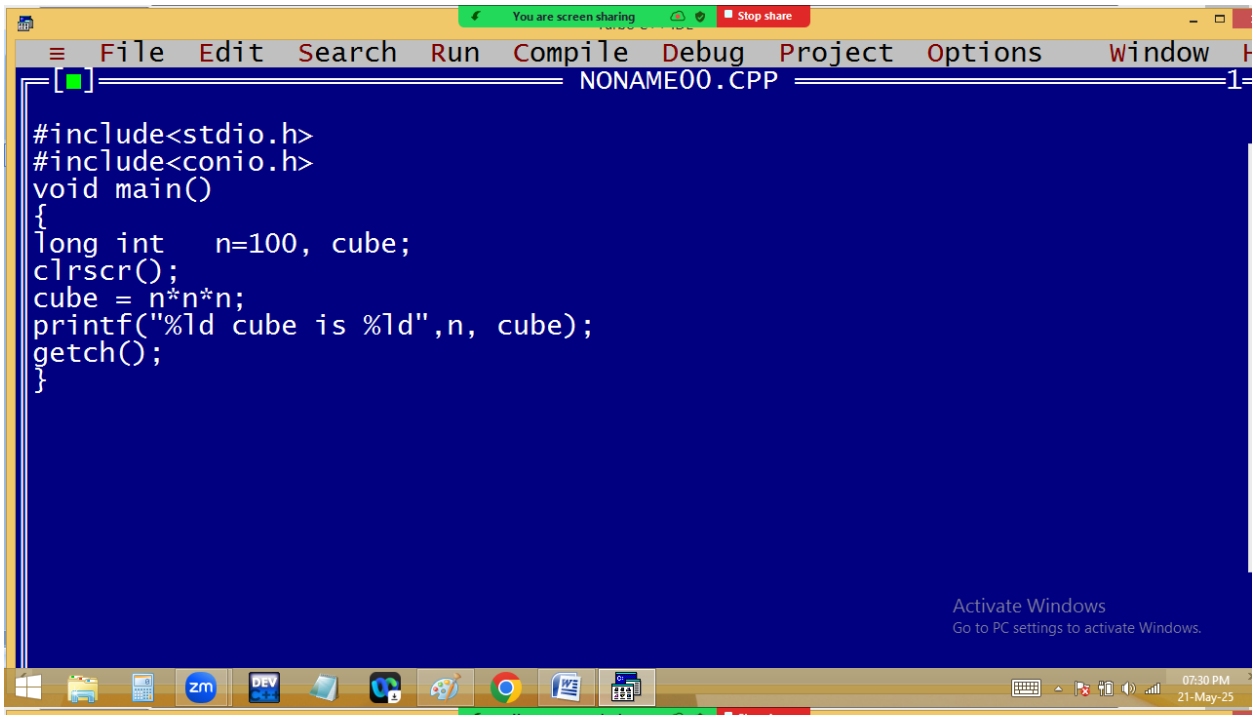
Finding cube value?

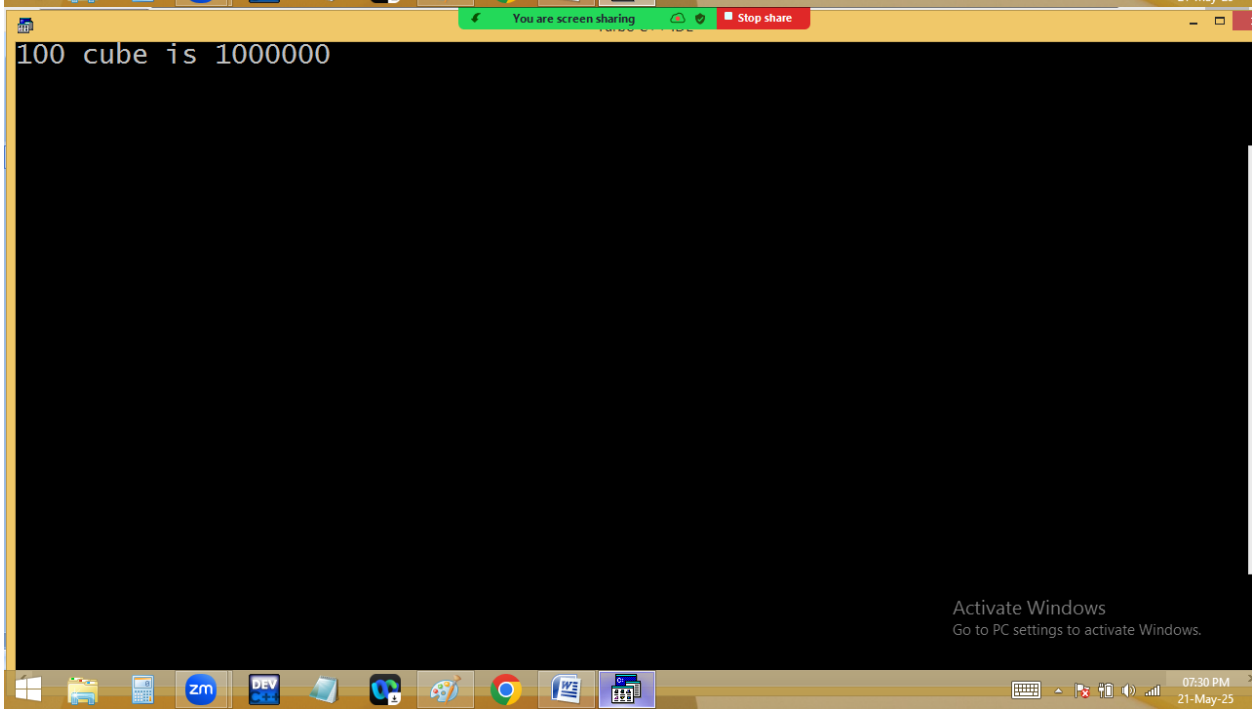

You are screen sharing
Stop share
File Edit Search Run Compile Debug Project Options Window
NONAME00.CPP
#include<stdio.h>
#include<conio.h>
void main()
{
int n=10,cube;
clrscr();
cube = n*n*n;
printf("%d cube is %d",n, cube);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
07:26 PM
21-May-25

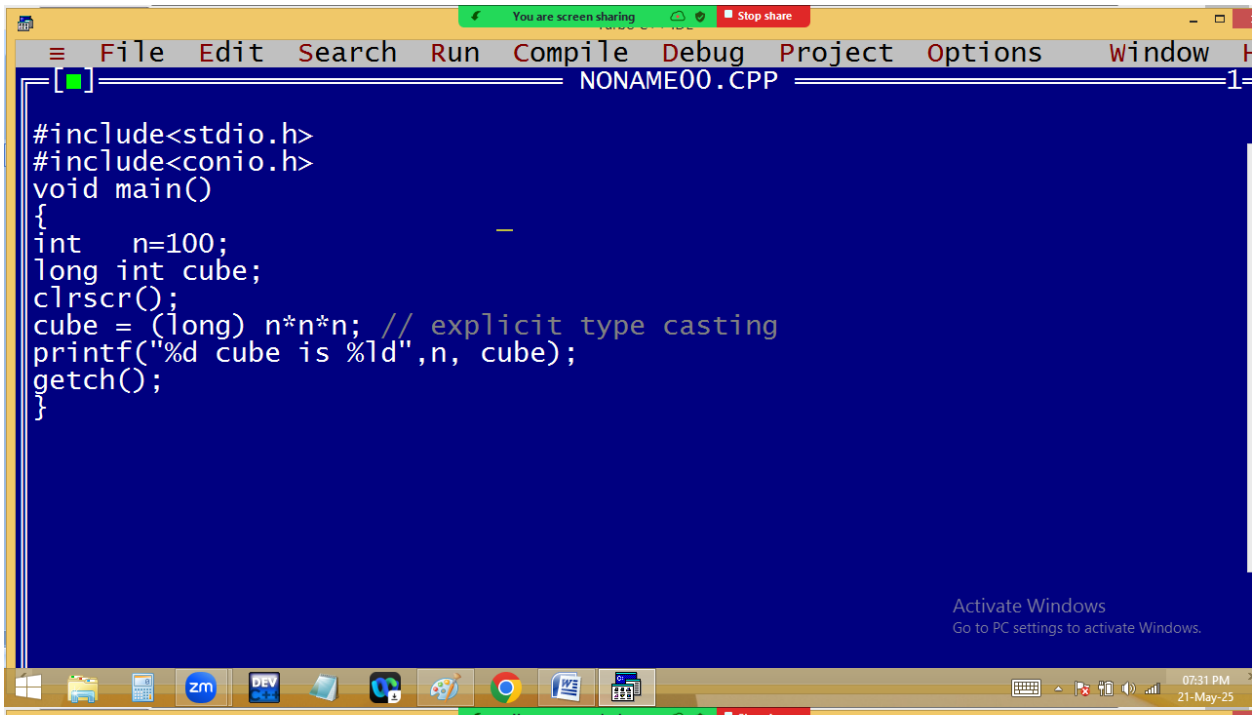

You are screen sharing
Stop share
10 cube is 1000
Activate Windows
Go to PC settings to activate Windows.
07:26 PM
21-May-25

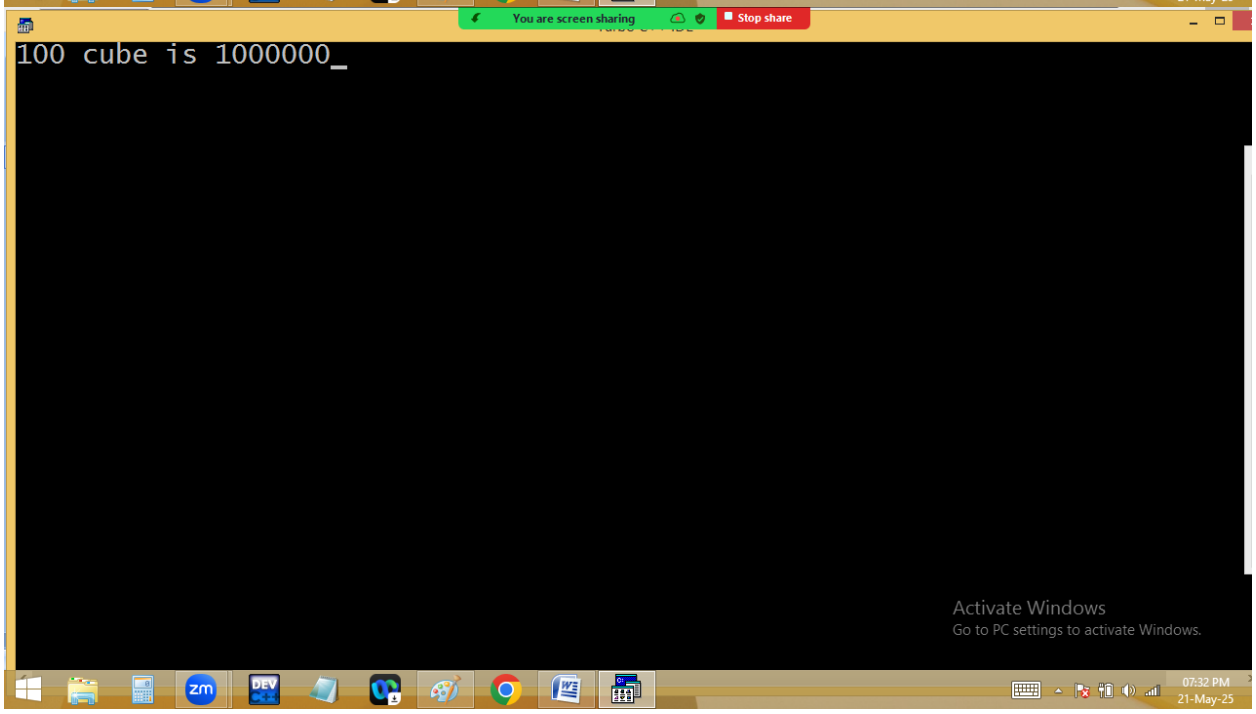

You are screen sharing
Stop share
File Edit Search Run Compile Debug Project Options Window
NONAME00.CPP
#include<stdio.h>
#include<conio.h>
void main()
{
int n=100,cube;
clrscr();
cube = n*n*n;
printf("%d cube is %d",n, cube);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
07:27 PM
21-May-25

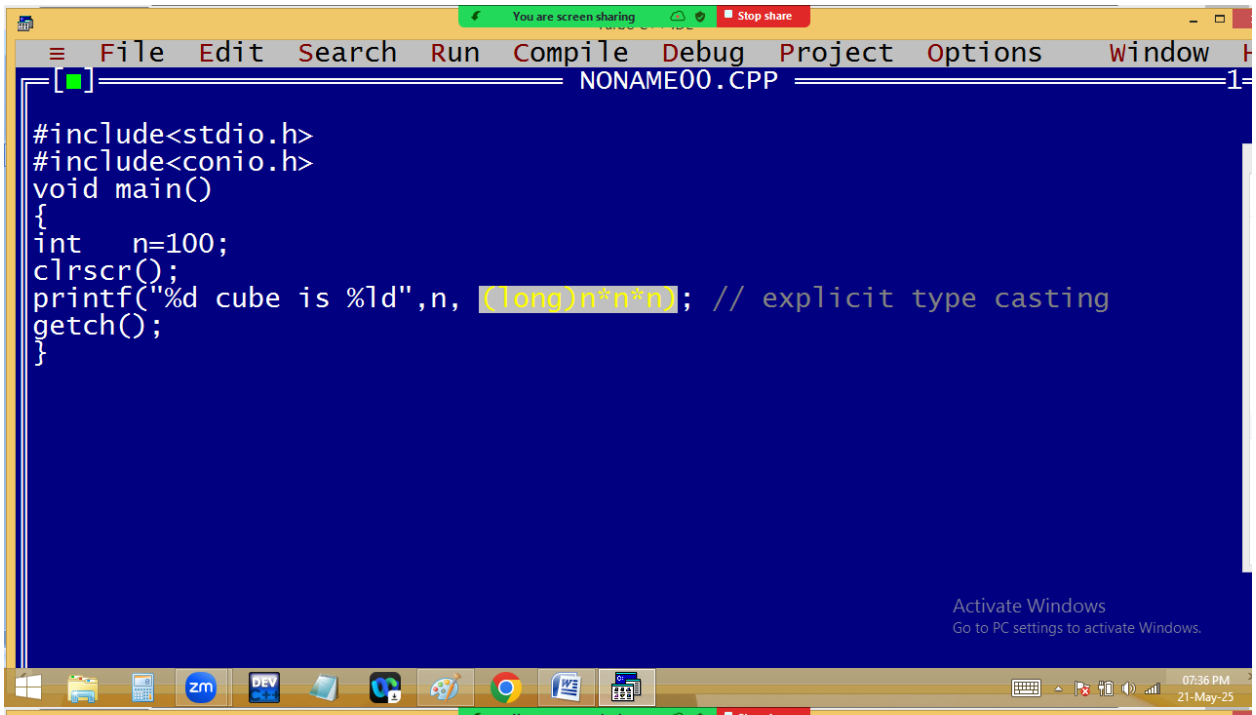

You are screen sharing
Stop share
100 cube is 16960
Activate Windows
Go to PC settings to activate Windows.
07:27 PM
21-May-25

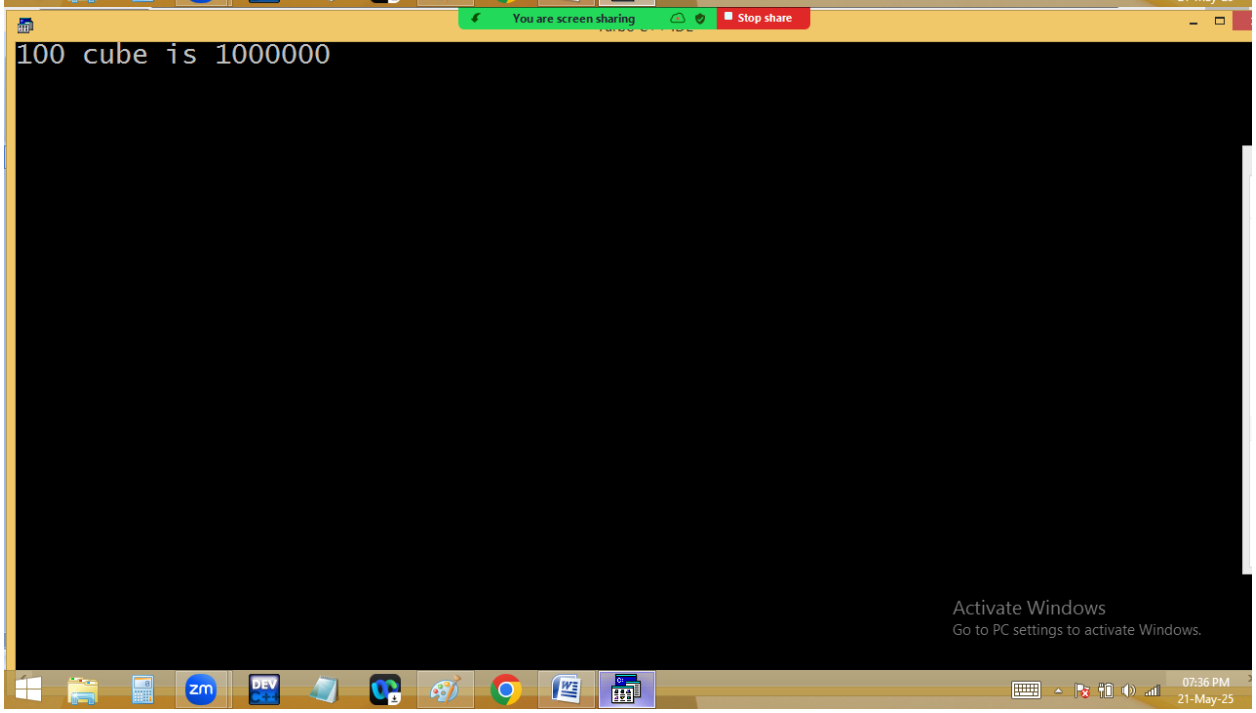

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File Edit Search Run Compile Debug Project Options Window
NONAME00.CPP
#include<stdio.h>
#include<conio.h>
void main()
{
long int n=100, cube;
clrscr();
cube = n*n*n;
printf("%ld cube is %ld",n, cube);
getch();
}
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100 cube is 1000000
Activate Windows
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NONAME00.CPP
#include<stdio.h>
#include<conio.h>
void main()
{
int n=100;
long int cube;
clrscr();
cube = (long) n*n*n; // explicit type casting
printf("%d cube is %ld",n, cube);
getch();
}
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100 cube is 1000000
Activate Windows
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NONAME00.CPP
#include<stdio.h>
#include<conio.h>
void main()
{
int n=100;
clrscr();
printf("%d cube is %ld",n, (long)n*n*n); // explicit type casting
getch();
}
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100 cube is 1000000
Activate Windows
Go to PC settings to activate Windows.
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